

Practical No.1: Identify various blocks of 8051 microcontroller development board.

I Practical Significance

Microcontroller has wide application in electronic system needing real time processing/control, starting from domestic application such as washing machine, TV and air conditioners. Microcontrollers are also used in automobiles, process control industries, cell phones, robotics and in space application. This practical will help the students to develop skills to Identify various blocks of 8051 development board.

II Relevant Program Outcomes (POs)

- **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunications engineering related problems.
- **Engineering tools:** Apply relevant Electronics and Telecommunications technologies and tools with an understanding of the limitations.
- **Individual and team work:** Function effectively as leader and team member in diverse/multidisciplinary teams.
- **Lifelong learning:** Engage in independent and life-long learning activities in the context of technological changes also in the Electronics and Telecommunication engineering and allied industry.

III Competency and Practical skills

This practical is expected to develop the following skills for the industry identified competency '**Maintain microcontroller-based systems**':

- Identify the components on microcontroller development board and understand their functions.
- Use integrate development environment (IDE).

IV Relevant Course Outcome(s)

- Analyze architecture of microcontroller ICs

V Practical Outcome

- Identify various blocks of 8051 microcontroller development board.

VI Relevant Affective domain related Outcome(s)

- Demonstrate working as a leader/a team member.
- Maintain tools and equipment.
- Follow ethical practices.

VII Minimum Theoretical Background

Microcontroller is a single chip microcomputer made through VLSI fabrication. 8051 is the microcontroller of the MCS-51 family introduced by Intel Corporation. It has inbuilt components such as CPU, internal RAM and ROM, timers/counters, serial ports, interrupts and I/O ports.

Resources used

Sr. No.	Instrument /Components	Specification	Quantity
1.	Micro controller kit	Single board 8k RAM 16x2 LCD Display.	1
2.	Desktop PC.	loaded with open Source IDE	1

Actual procedure followed (use blank sheet provided if space is not sufficient)

- 1) Start Keil software by double clicking on Keil Icon.
- 2) To create new project, click on project & select new project.
- 3) Select appropriate location for new project & type project name, click on same button.
- 4) "Select device for target target-1" window will open. It displays a list of manufacturer of microcontroller.
- 5) Double click on ATMEL or INTEL, list of supported microcontroller get displays.
- 6) click file pull down menu, select new, a text editor window will open, save the file in same folder where project was stored. Give extension .ASM or .AS1.

III Precautions followed (use blank sheet provided if space not sufficient)

- 1) Do not power up development board when identifying block.

IV Observations (use blank sheet provided if space not sufficient)

Observe development board and list various components and write their functions.

Sr. No.	Component	Function
1	8051	
2	crystal oscillator	
3	capacitor	
4	Regulator IC	
5	Reset switch	

- Result Identification of various block of 8051 development board
- 1) 8051 development board integrated on board two VART, keypad on ADC I/P (2) 4 pin-2, if sockets
 - 3) 4x4 (4) Stepper motor interface matrix keypad.
 - (5) Traffic light controller modules.

XVI Interpretation of Results (Hardware and software features of 8051 development board and Keil)

At vision for window is a integrated development environment that combined project management source code editing & program debugging in are single.

XVII Conclusions and Recommendations

When starting a new project simply select a controller to use from the device data base & u vision IDE sets all compiler assembly & memory.

XVIII Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design more such questions so as to ensure the achievement of identified CO.

1. List internal and external interrupts available to 8051 microcontroller?
2. List any four features of Keil IDE.
3. List MCS51 family members and write their specifications.
4. Explain the selection criteria for selecting a microcontroller device.
5. List I/O ports available in 8051 microcontroller.

[Space for Answers]

1) → Timer 0 overflow intercept - IFO
 Timer 1 overflow intercept - IF1
 external hardware intercept - INTO
 External hardware intercept - INT1
 serial communication intercept - RI/TI

2) → i) The u vision device database automatic configure development tools target microcontroller

3) → The u vision IDE integrates additional

third party tool. Third party tools device programming.

iii) μ vision in corporates project manage editor, debugger in a single enrollment.

iv) The μ vision simulator is the only debugger that completely simulated all on chip peripherals of the ATME1 philips, Samsung.

v) 8051 has have four I/O ports each of 8-bit which can be configure as I/O or O/P.

1) Pin configuration.

2) Input pin.

3) port 0.

4) $\rightarrow$ steps for selection criteria of 8051 micro controller.

Step 1:- Make alist of required hardware interface.

Step 2:- Examines the software architecture.

Step 3:- select the architecture.

Step 4:- Identify memory needs.

Step 5:- start searching for microcontroller.

Step 6:- Examines costs & power constant.

Step 7:- check port availability.

Step 8:- select a development kit.

Step 9:- Investigate compilers & port.

Step 10:- start experimenting.

Practical No. 2: Write sample assembly language program using various addressing modes and assembler directives

I Practical Significance

The addressing modes specifies the way data can be moved or copied from source to destination location thus providing various options and flexibility for data transfer. This practical will help the students to develop skills to allow the programmer to write structured program which is essential to code maintainability.

II Relevant Program Outcomes (POs)

- **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunications engineering related problems.
- **Experiments and practice:** Plan to perform experiments and practices to use the results to solve broad-based Electronics and Telecommunication engineering problems.
- **Lifelong learning:** Engage in independent and life-long learning activities in the context of technological changes also in the Electronics and Telecommunication engineering and allied industry.

III Competency and Practical Skills

This practical is expected to develop the following skills for the industry identified competency 'Maintain microcontroller-based systems':

- Use of addressing modes and how it provides flexibility in software development.
- Use of directives to instruct the assembler.
- Develop assembly programs for microcontroller.

IV Relevant Course Outcome(s)

- Interpret the program for 8051 in assembly language for the given operations.

V Practical Outcome

- Write sample assembly language program using various addressing modes and assembler directives.

VI Relevant Affective domain related Outcome(s)

1. Follow ethical practices.

VII Minimum Theoretical Background

Addressing Modes: A microcontroller provides various methods for accessing data needed in the execution of an instruction. The various methods of accessing the data are called addressing modes.

1. **Immediate addressing mode:** The data is provided in instruction itself. The data is identified by #before the numerical value
Ex: MOV A, #05H (The immediate data 05h provided in instruction is moved into A register).

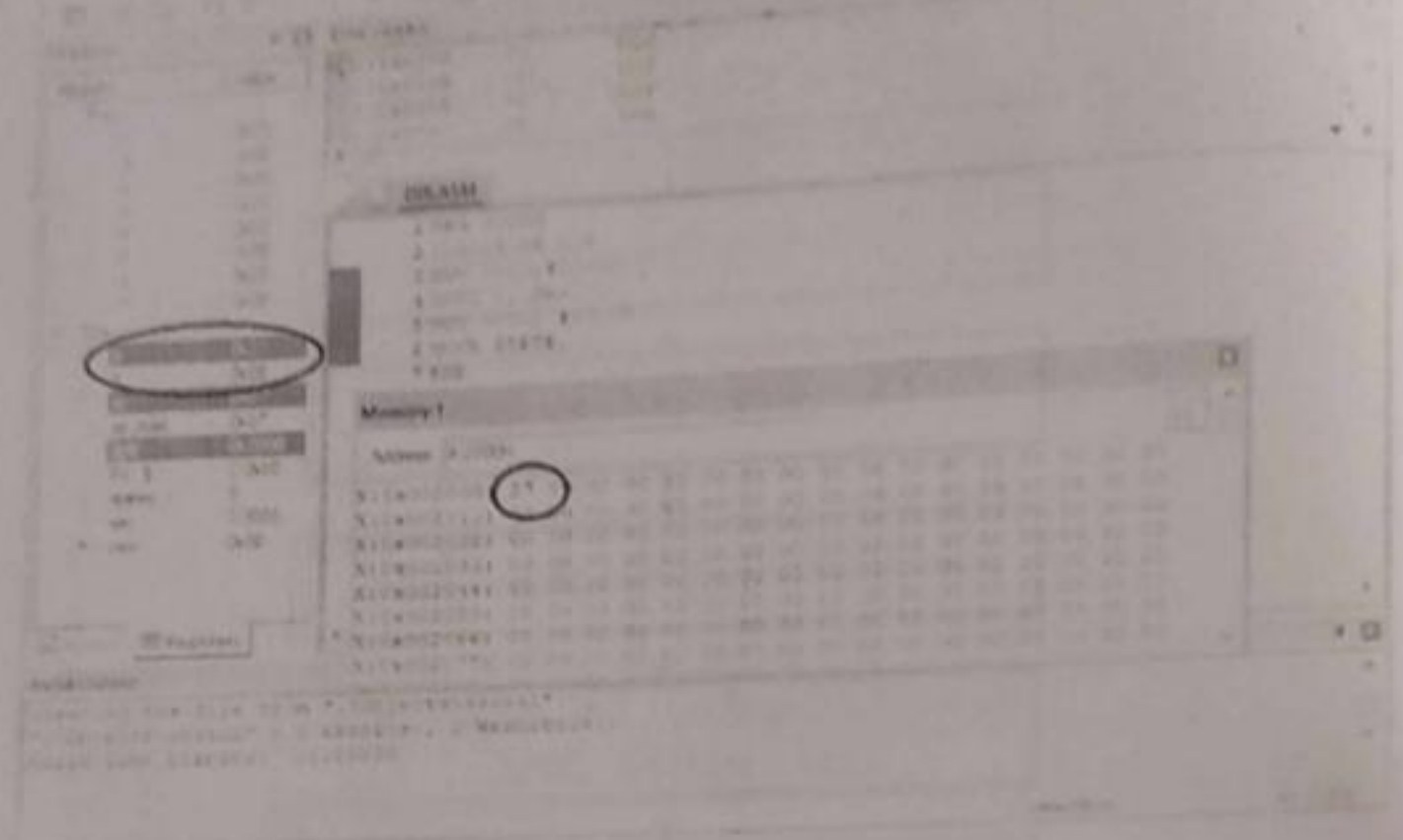


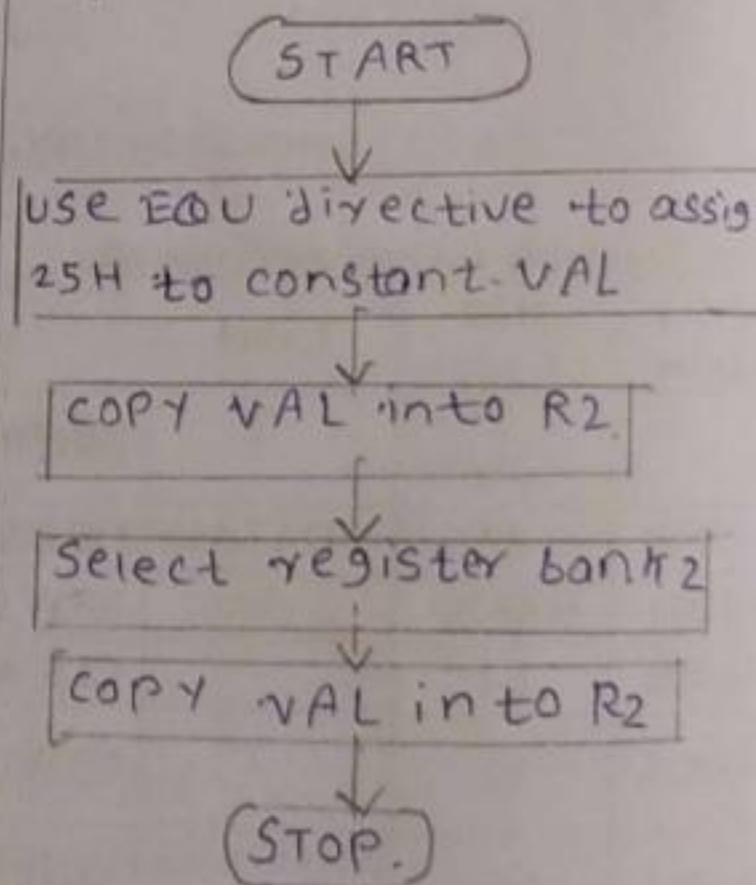
Fig 2.4 Output window

Problem statement #1 for student: Write a program to transfer the data 25h into register R Bank 0 and register R2 of Bank 1 using immediate and register addressing mode.

Step 1-Algorithm

- 1) Start
- 2) copy 25H into Reg A
- 3) copy Reg A into R2
- 4) select bank 2
- 5) copy Reg A into R2.
- 6) stop.

Step 2-Flowchart



// program to transfer the data 25 h into
// register R2 of Bank 0 & register
// R2 of Bank 1 using immediate & register addressing
mode

ORG 0000H

MOV A, #25H // copy 25H into Reg A.

MOV R2, A // copy Reg A into R2.

SETB PSW, 4 // select Register Bank 2.

MOV R2, A // copy Reg A into R2.

END

Assembly Language Program

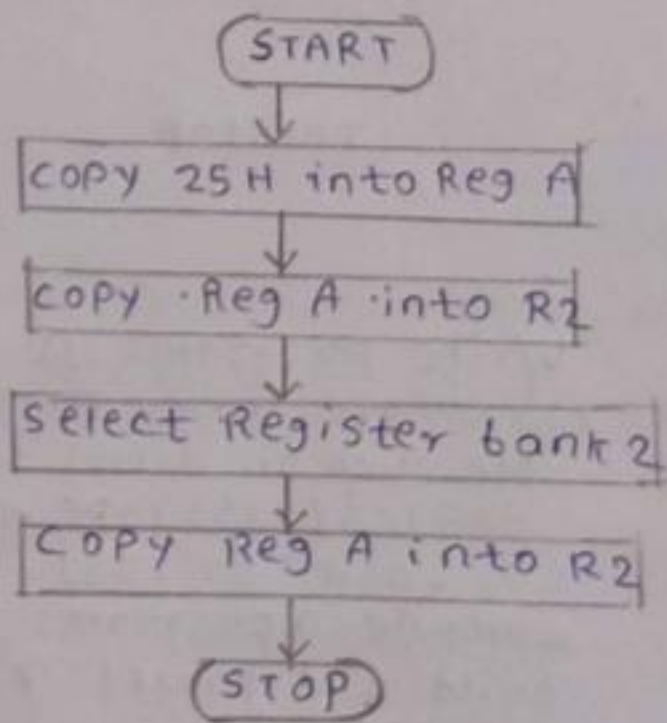
Memory Address	Hex Code	Label	Mnemonics	Comments
			ORG 0000H	
0000H	7425		MOV A, #25H	copy 25H into Reg A
0002H	FA		MOV R2, A	copy Reg A into R2
0003H	D2D4		SETB PSW.4	Select register bank 2
0005H	FA		MOV R2, A	copy Reg A into R2
			END	stop

Problem statement #2 for student: Write a program using EQU directive.

Step 1-Algorithm

- 1) start.
- 2) Use EQU directive to assign 25H to constant VAL.
- 3) copy VAL into R2.
- 4) select bank 2
- 5) copy VAL into R2
- 6) stop.

Step 2-Flowchart



Step 3: Assembly Language Program

Memory Address	Hex Code	Label	Mnemonics	Comments
			ORG 0000H	
			VALUE EQU 25H	Assign 25H to constant
0000H	7A25		MOV R2, #VAL	copy VAL into R2
0002H	D2D4		SETB PSW.4	select register bank
0004H	7A25		MOV R2, #VAL	copy VAL into R2
			END	Stop

XI Resources Used

S. No.	Instrument /Components	Specification	Quantity
1	Desktop	loaded with IDE Keil software.	1

- XII Actual Procedure Followed (use blank sheet provided if space not sufficient)
- Write Program: (1) Start Keil by double clicking on Keil icon (2) create a new project (3) select device for target (4) double click on ATMEGA or INTEL & select 80C51AH or AT89C51. (5) Type the program in text editor & save as .asm or .a51.
- Compile a program: (6) Right click on source group & build the target (7) check for any errors in the output window & remove if any.
- Run, Debug the program: (8) click on Debug & start simulation & start/stop debug session (9) Run step by step (10) observe o/p on the project display (11) Note the contents
- XIII Precautions Followed (use blank sheet provided if space not sufficient)
1. Check rules/syntax of assembly programming.

: 0x00:	00	00	25	00	00	00	00
: 0x08:	00	00	00	00	00	00	00
: 0x10:	00	00	25	00	00	00	00
: 0x18:	00	00	00	00	00	00	00
: 0x20:	00	00	00	00	00	00	00
: 0x28:	00	00	00	00	00	00	00
: 0x30:	00	00	00	00	00	00	00

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Address : 1 = 00H

I:	0x00 :	00	00	00	00	00	00	00
I:	0x08 :	00	00	00	00	00	00	00
I:	0x10 :	00	00	00	00	00	00	00
I:	0x18 :	00	00	00	00	00	00	00
I:	0x20 :	00	00	00	00	00	00	00
I:	0x28 :	00	00	00	00	00	00	00
I:	0x30 :	00	00	00	00	00	00	00

I:	0x00 :	00	00	25	00	00	00	00	00
I:	0x08 :	00	00	00	00	00	00	00	00
I:	0x10 :	00	00	25	00	00	00	00	00
I:	0x18 :	00	00	00	00	00	00	00	00
I:	0x20 :	00	00	00	00	00	00	00	00
I:	0x28 :	00	00	00	00	00	00	00	00
I:	0x30 :	00	00	00	00	00	00	00	00

Result:-

Address : 1:00H

13 0000 00 00 00 00 00 00 00

I: 0x08: 00 00 00 00 00 00 00 00

I: 0x10: 00 00 00 00 00 00 00 00

I: 0x18: 00 00 00 00 00 00 00 00

I: 0x20: 00 00 00 00 00 00 00 00 00

1: 0X281 00 00 00 00 00 00 00 00

I: 0X30: 00 00 00 00 00 00 00 00 00

Accumulator	
R1	
RAM memory location	

Accumulator	
DPTR	
2000H memory location	

Results (Output of the Program)

Interpretation of Results (Give meaning of the above obtained results)

- Copied 25H into register A & after copied in R1.
- By using EQU directive we assign 25H to constant VAL & copied VAL into R2.

II Conclusions and Recommendation

By performing this practical we learnt disassembly language, add testing modes & assembler directive.

III Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design more such questions so as to ensure the achievement of identified CO

- State significance of the symbols @ used in addressing mode.
- Develop a program to copy the value 55H into RAM memory locations 40H to 41H using (a) direct addressing mode, (b) register indirect addressing mode without a loop, and (c) with a loop.
- State the contents of each ROM location for the following data

ORG 2000H

DATA1: DB "Earth"

DATA2: DB "987-65"

DATA3: DB "GABEH 98"

- Copy the byte in Accumulator to register R2 using at least three different addressing modes.

[Space for Answers]

- 1) The addressing mode specifies a rule for interpreting or modifying the address field of the instruction before the operand is actually executed.

Practical No. 3: Write an assembly language program (ALP) to perform arithmetic operations: addition, subtraction, multiplication and division.

I Practical Significance

8051 microcontroller has single instruction arithmetic operations. Applications such as BCD and ASCII conversions and checksum byte testing require arithmetic operations. This practical will help the students to develop skills to write assembly program for arithmetic operations.

II Relevant Program Outcomes (POs)

- **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunications engineering related problems.
- **Experiments and practice:** Plan to perform experiments and practices to use the results to solve broad-based Electronics and Telecommunication engineering problems.
- **Engineering tools:** Apply relevant Electronics and Telecommunication technologies and tools with an understanding of the limitations.

III Competency and Practical Skills

This practical is expected to develop the following skills for the industry identified competency 'Maintain microcontroller based systems':

- Develop the code for the given problem.
- Debug the code and rectify the errors.
- Run the code.

IV Relevant Course Outcome(s)

- Interpret the program for 8051 in assembly language for the given operations.

V Practical Outcome

- Write an assembly language program (ALP) to perform arithmetic operations: addition, subtraction, multiplication and division.

VI Relevant Affective domain related Outcome(s)

- Follow ethical practices.

VII Minimum Theoretical Background

Accumulator is an 8 bit register and is used for all arithmetic and logic operations. To perform arithmetic operations, it is necessary that one of the two operands should be in accumulator. Register B is used for multiplication and division purpose. There are four register banks available in 8051; each register bank consists of 8 registers (R0 to R7).

// program to perform series addition of ten bytes &
 // store the lower byte of result in 30h & higher
 // byte of result in 31h.

ORG 0000H

CLR A // clear reg A ^{and} carry.

CLRC

MOV R2, #0AH // Initialize R2 as counter.

MOV RO, #50H // Initialize RO as memory pointer.

UP: ADD A, @RO // Add the data pointed by RO to Acc

JNC NEXT // If carry is generated.

INC 30H // update higher byte of result.

NEXT: MOV 31H, A // store lower byte of result to memory location 31H

INC RO // Increment RO to point to the next data.

DJNZ R2, UP // counter = counter - 1, If it is zero then goto

END // UP

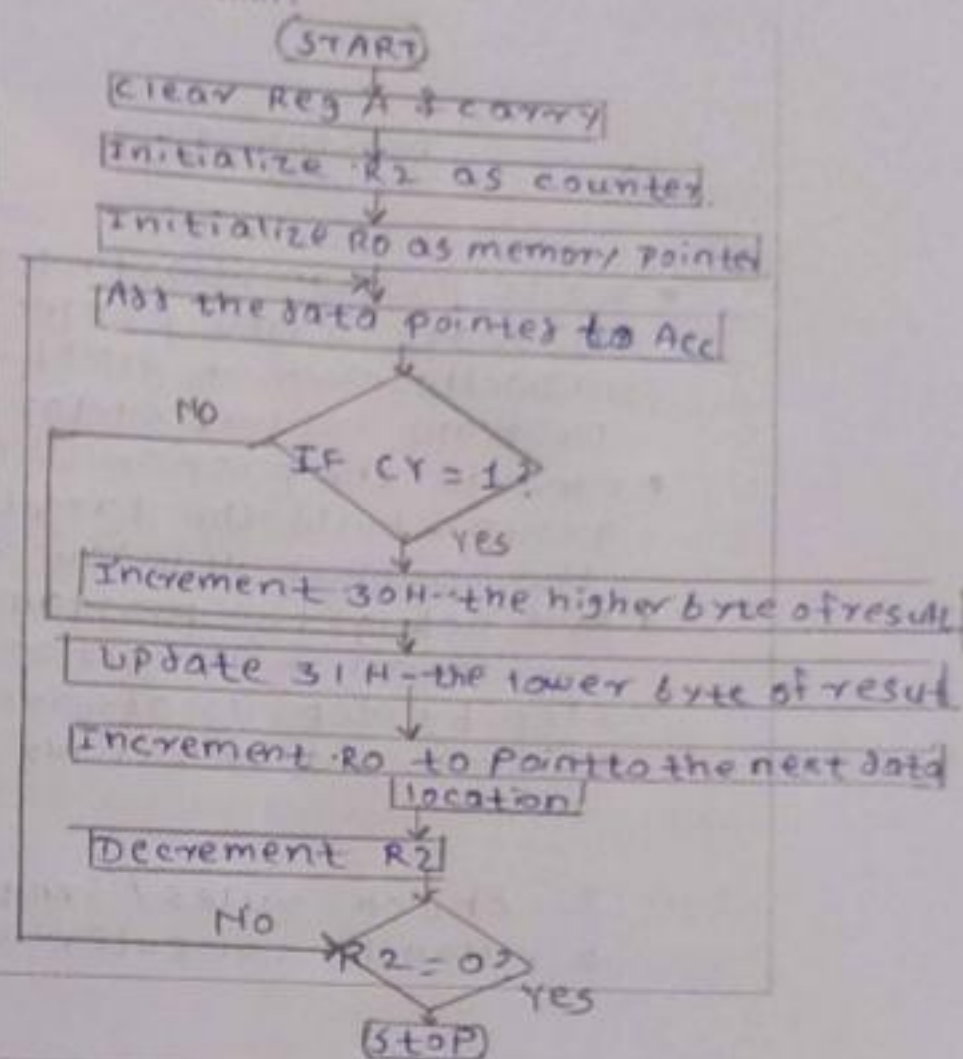
Memory Address	Hex Code	Label	Mnemonics	Comments
			ORG 0000H	
0000H	E4		CLR A	clear reg A.
0001H	C3		CLRC	clear carry.
0002H	7A0A		MOV R2, #0AH	initialize R2 as counter
0004H	7850		MOV RO, #50H	initialize RO as memory pointer.
0006H	26	UP:	ADD A, @RO	Add number from memory with A
0007H	5002		JNC NEXT	store lower byte to memory location
0009H	0530		INC 30H	if carry is generated 31H.
				update higher byte of result.
000BH	F531	NEXT:	MOV 31H, A	increment pointer reg
				for next number
000DH	08		INC RO	counter = counter - 1,
				If it is zero then goto up
000EH	DAF6		DJNZ R2, UP	stay here
			END	

Problem statement: Write a program to perform series addition of 5 numbers and store the lower byte of result in 30H and higher byte of result in 31H.

Step 1 - Algorithm

- 1) start.
- 2) clear reg A & carry.
- 3) initialize R2 as counter.
- 4) initialize R0 as memory pointer.
- 5) Add the data pointed by R0 to Acc.
- 6) if carry is not generated go to Step 8.
- 7) increment 30H - the higher byte of result.
- 8) update 31H - the lower byte of result.
- 9) increment R0 to point to the next data.
- 10) decrement R2.
- 11) if R2 is not zero then go to step 5.
- 12) stop.

Step 2 - Flowchart



Step 3 - Assembly Language Program

Memory Address	Hex Code	Label	Mnemonics	Comments
			ORG 0000H	
0000H	E4		CLRA	clear reg A
0001H	C3		CLRC	clear carry
0002H	7A0A		MOV R2, #0AH	initialize R2 as counter
0004H	7850		MOV R0, #50H	initialize R0 as memory pointer
0006H	26	UP:	ADD A, @R0	Add number from memory with A
0007H	5002		JNC NEXT	store lower byte to memory location 31H
0009H	0530		INC 30H	if carry is generated update higher byte of result
000BH	F531	NEXT:	MOV 31H, A	increment pointer reg for next number
000DH	08		INC R0	counter = counter - 1, if it is zero the go to up
000EH	DAF6		DJNZ R2, UP	stay here
			END	

XI Resources Used

S. No.	Instrument / Components	Specification	Quantity
1	Desktop PC	Loaded with open source IDE	1

XII Actual Procedure Followed (use blank sheet provided if space not sufficient)

- Write program: (1) Start Keil by double clicking on Keil icon. (2) Create a new project. (3) Select device for target. (4) Double click on AT89C51. (5) Type the program in text editor & save as filename.asm extension.
- Compile the program: (6) Right click on Debug Source group & build the target. (7) Check for any errors in the output window & remove if any.
- Run, Debug the program: (8) Click on Debug & start simulation & start/stop debug session. (9) Run the program step by step. (10) Observe the output. (11) Note the value of the result of various operation in observation table.

XIII Precautions Followed (use blank sheet provided if space not sufficient)

1. Check rules / Syntax of assembly programming.
2. Handle computer system carefully.

XIV Observations for sample program (use blank sheet provided if space not sufficient)

Sr. No.	Register used in the code	Operation	Result after execution
1	R0	Addition	5H
2	R1	Subtraction	5H
3	R2	Multiplication (LSB)	0CH
4	R3	Multiplication (MSB)	00H
5	R4	Division (Quotient)	09H
6	R5	Division (Remainder)	05H

XIX Results (Output of the Program)

Result of 5 byte (01+01+01+01+01) is 5H is stored in 3AH memory location & 00H & 31H (result in 8 bit).

XX Interpretation of Results (Give meaning of the above obtained results)

Various operation can be performed & result is in 8 bit register by using indirect addressing mode. No store in memory in access assume no. is store in memory location.

Address D: 00H

D: 0x10 : 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

D: 0x20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

D: 0A30:02 F7 00 00 00 00 00 00 00 00 00 00 00 00

D: 0x40: 00 00 00 00 00 00 00 00 00 00 00 00 00

D : 0x50 : 04 11 02 15 15 01 ED 9A FA 00 00 00

D : 0X60 : 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

D: 0x70: 00 00 00 00 00 00 00 00 00 00 00 00 00

Result:

Address: D: 00H

D: 0x10 : 00 00 00 00 00 00 00 00 00 00 00 00

D: 0x20 % 00 00 00 00 00 00 00 00 00 00 00 00

D: 0x30 : 00 00 00 00 00 00 00 00 00 00 00 00

D: 0x40 : 00 00 00 00 00 00 00 00 00 00 00 00

D: 0x50 : 04 11 02 05 15 01 ED 9A 44 FA 00

D: 0x60 ; 00 00 00 00 00 00 00 00 00 00 00 00

D: DX 70 : 00 00 00 00 00 00 00 00 00 00 00

By using various instructions we can performed different arithmetic operations.

Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design more such questions so as to ensure the achievement of identified CO

1. If CY=1, A=95H, and B=4FH prior to the execution of "SUBB A, B", what will be the contents of A after the subtraction?
2. Is this a valid 8051 instruction? "DIV A, R1". Explain your answer.
3. Write the status of the CY, AC and P flag after the addition of 9CH and 64H in the following instructions.
MOV A, #9CH
ADD A, #64H

[Space for Answers]

1) → set B C A # 95H
B, # 4FH.

SUBB A, 8A = 55h 4FH & carry = 46h - 1 = 45H.

In multiplication of two byte in the 8051 we must register A & register B. MATLAB to MATLAB: $A \times B = AB$ (16 bits). In this valid 8051 mul A, R1. the source address of multiplication operation must be in A & B. 12 in byte.

- 2) → instructions :- i) ACALL - Absolute call
ii) ADD, ADDC, → Add Accumulator.
iii) AJMP - Absolute jump.
iv) ANL Bitwise AND DIV R1 → Divide the unsigned value of the 'B' register. The resulting quotient is replaced in the accumulator & the remainder is placed in the B register.

96	100	111	00
64	011	001	00
100	000	00000	

AC = 0

CY = 1

P = 1

Practical No. 4: Write an ALP to transfer data from source to destination location of internal/external data memory

Practical Significance

8051 microcontroller is used in embedded system. Understanding its memory organization helps in making optimal use of internal RAM and ROM and for applications which require additional memory, the external memory can be accessed. This practical will help the students to develop skills to transfer data from source to destination location.

Relevant Program Outcomes (POs)

- **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunications engineering related problems.
- **Experiments and practice:** Plan to perform experiments and practices to use the results to solve broad-based Electronics and Telecommunication engineering problems.
- **Engineering tools:** Apply relevant Electronics and Telecommunications technologies and tools with an understanding of the limitations.

Competency and Practical Skills

This practical is expected to develop the following skills for the industry identified competency 'Maintain microcontroller based systems':

- To use different instructions to transfer, exchange contents of blocks.

Relevant Course Outcome(s)

- Interpret the program for 8051 in assembly language for the given operations.

Practical Outcome

- Write an ALP to transfer data from source to destination location of internal/external data memory.

Relevant Affective domain related Outcome(s)

- Follow ethical practices

Minimum Theoretical Background

Block Transfer: In block transfer the content of source memory location are transferred to destination memory location. Block transfer is done from internal RAM memory location to internal RAM memory location or from internal RAM memory location to external RAM memory location.

External Memory: 8051 microcontroller has 128 bytes of internal RAM memory (data memory) and 4 Kbytes of internal ROM memory (program memory). In addition to internal memory 64 Kbytes of data memory and 64 Kbytes of program can be interfaced with 8051 microcontroller.

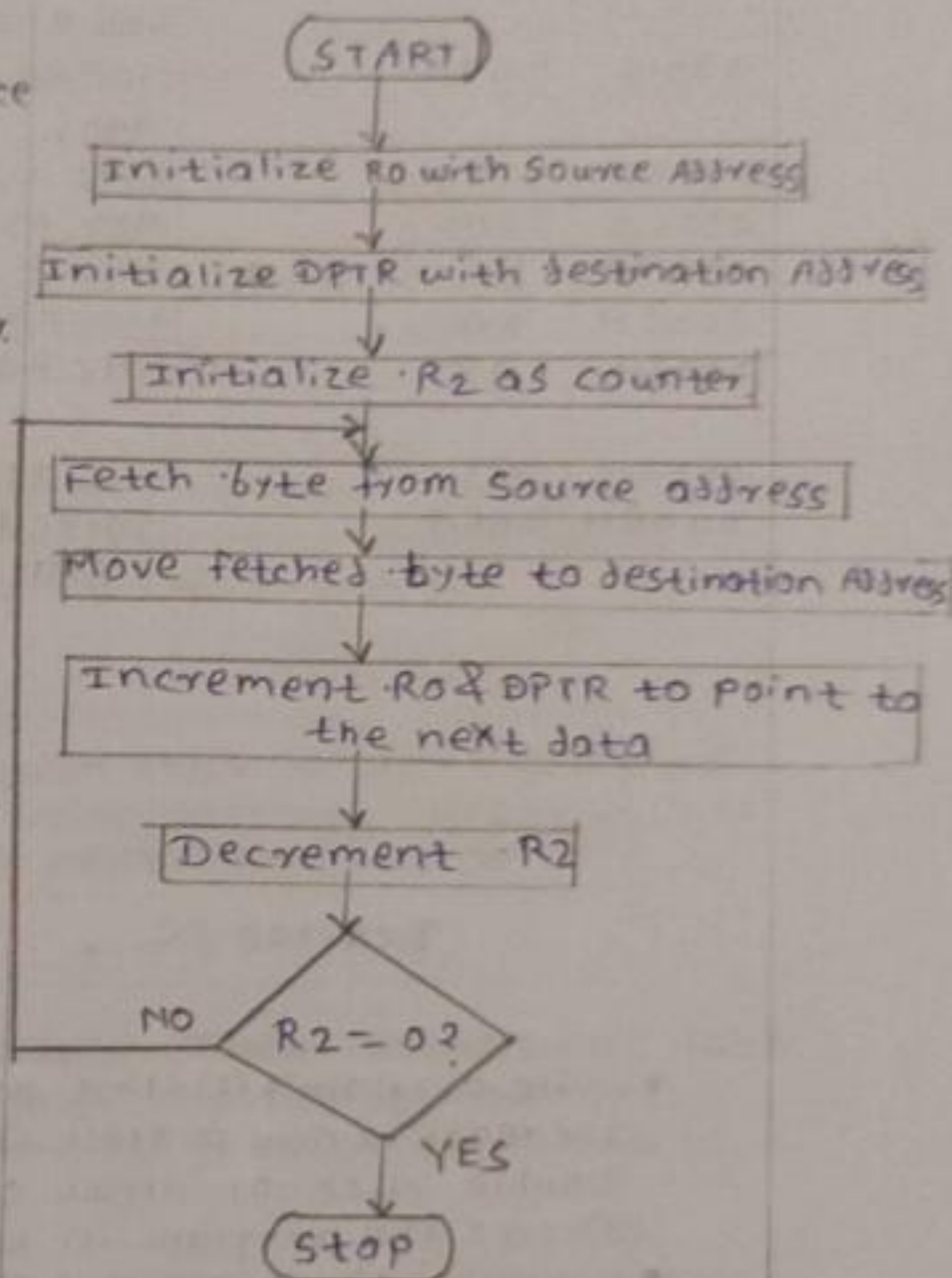
To interface external memory to 8051 EA^{*} pin should be connected to ground.

Problem statement 1 for student: Write a program to transfer a block of 100 bytes from memory location 40H to external memory location 2000H.

Step 1: Algorithm

- 1) start
- 2) Initialize R0 with source address.
- 3) Initialize DPTR with destination address.
- 4) Initialize R2 as counter.
- 5) Fetch byte from source address.
- 6) move fetched byte to destination address.
- 7) Increment R0 & DPTR to point to the next data.
- 8) Decrement R2 & if it is not zero then go to step 5.
- 9) stop.

Step 2: Flowchart



Step 3: Assembly Language Program Code

Memory Address	Hex Code	Label	Mnemonics	Comments
000H	7480		ORG 0000H MOV R0, #40H MOV	Source Address Destination - Address.
0002H	902000		DPTR, #2000H	No. of bytes to be moved
0005H	7A05		MOV R2, #05	Fetch byte from source
0007H	E6	BACK:	MOV A, @R0	Store to destination
0008H	F0		MOVX @DPTR, A	Increment R0 & DPTR
0009H	08		INC R0	To point to the next location
000AH	A3		INC DPTR	
000BH	DAFA		DJNZ R2, BACK	Repeat till all data transfer
			END	

XI Resources Used

Sr. No.	Name of Resource.	Specification	Quantity
1	Desktop PC	Loaded with open source IDE	1

XII Actual Procedure Followed (use blank sheet provided if space not sufficient)

- write program: (1) Start Keil by double clicking on Keil icon.
(2) create a new project. (3) select device for target. (4) Double click on ATME1 or INTEL & select 80C51AH or AT89C51AH.
(5) type the program in text editor & save as .asm or .a51
- compile the program: (6) Right click on source group & build the target. (7) check for any errors in the output window & remove if any.
- Run & Debug the program: (8) click on Debug & start simulation & start/stop debug session.
(9) Run the program step by step.
(10) observe the output on the project window. It will display all internal registers of 8051 & their contents.

XIII Precautions Followed (use blank sheet provided if space not sufficient)

- 1. check rules/syntax of assembly programming.

XIV Observations for problem statement (Use blank sheet provided if space is not sufficient)

Before execution		After execution	
Memory location	Data	Memory location	Data
40H	01	2000H	01
41H	02	2001H	02
42H	03	2002H	03
43H	04	2003H	04
44H	05	2004H	05

XV Results (Output of the Program)

The blocks of internal memory location 40H are transferred to the external memory location 2000H onwards.

XVI Interpretation of Results (Give meaning of the above obtained results)

We select five internal memory location from 40H & their respective data after execution the data blocks of 40H memory location are transferred to the external memory.

XVII Conclusions and Recommendation

Using indirect addressing mode blocks of data is transferred from internal memory location 20H to external memory location 2000H.

XVIII Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design more such questions so as to ensure the achievement of identified CO.

1. Give syntax of two instructions used in program to exchange data.
2. Write a program to exchange higher nibble and lower nibble of data 48H.
3. Write the contents of register A after execution of the following instructions
MOV A, #28H.
RRA
RRA
END

[Space for Answers]

Q → i) MOV A, @R0

MOV copied the value of vertical into operand 1

ii) MOV X DPTR, A :-
MOV X moved a byte to or
from external memory into from the
Accumulator.

2) → program to change lower nibble.

```
MOV DPTR, #114FH
MOV X A, @DPTR.
MOV B, A
INC DPL
MOV X A, @DPTR.
MOV R0, #0101
XCHG A, @R0
MOV X @DPTR, A
MOV A, B
DEC DPL
MOV X @DPTR.
```

3) → MOV A, #28H

RR A

RR A

END

The RR instruction shifts the bits of
the accumulator to the right.

MOV A, #20H ; A = 0010 1000

RR A ; A = 0001 0100

A = 1000 1010

content of the A after execution will
be 118AH.

Practical No. 5: Write ALP to find smallest / largest number from the given data bytes stored in internal/external data memory locations.

I Practical Significance

CJNE and DJNZ instructions of 8051 microcontroller perform two tasks in a single instruction. The application of these instructions can be demonstrated in finding smallest and largest numbers from a block of numbers. This practical will help students to develop skills to use the compare and loop instructions.

II Relevant Program Outcomes (POs)

- **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunications engineering related problems.
- **Experiments and practice:** Plan to perform experiments and practices to use results to solve broad-based Electronics and Telecommunication engineering problems.
- **Engineering tools:** Apply relevant Electronics and Telecommunication technologies and tools with an understanding of the limitations.

III Competency and Practical Skills

This practical is expected to develop the following skills for the industry identified competency: 'Maintain microcontroller based systems':

- Develop the code for the given problem.
- Debug the code and rectify the errors.
- Run the code
- Use of application of compare and loop instructions.

IV Relevant Course Outcome(s)

- Interpret the program for 8051 in assembly language for the given operations.

V Practical Outcome

- Write ALP to find smallest/largest number from the given data bytes stored in internal/external data memory locations

VI Relevant Affective domain related Outcome(s)

- Follow ethical practices

VII Minimum Theoretical Background

Largest number or smallest number can be found from a block of data using the compare and loop instructions. 8051 microcontroller is an example of CISC machine which has few complex instructions like CJNE and DJNZ.

CJNE destination, source, relative address.

In 8051, the actions of comparing and jumping are combined into a single instruction called CJNE. It compares two operands and jump to relative address if they are not equal. In addition, it changes the CY flag to indicate if the destination operand is larger/smaller. This instruction is used to find the largest/smallest number.

DJNZ Rn, relative

This instruction decrements the contents of register and jump to the relative address if not zero.

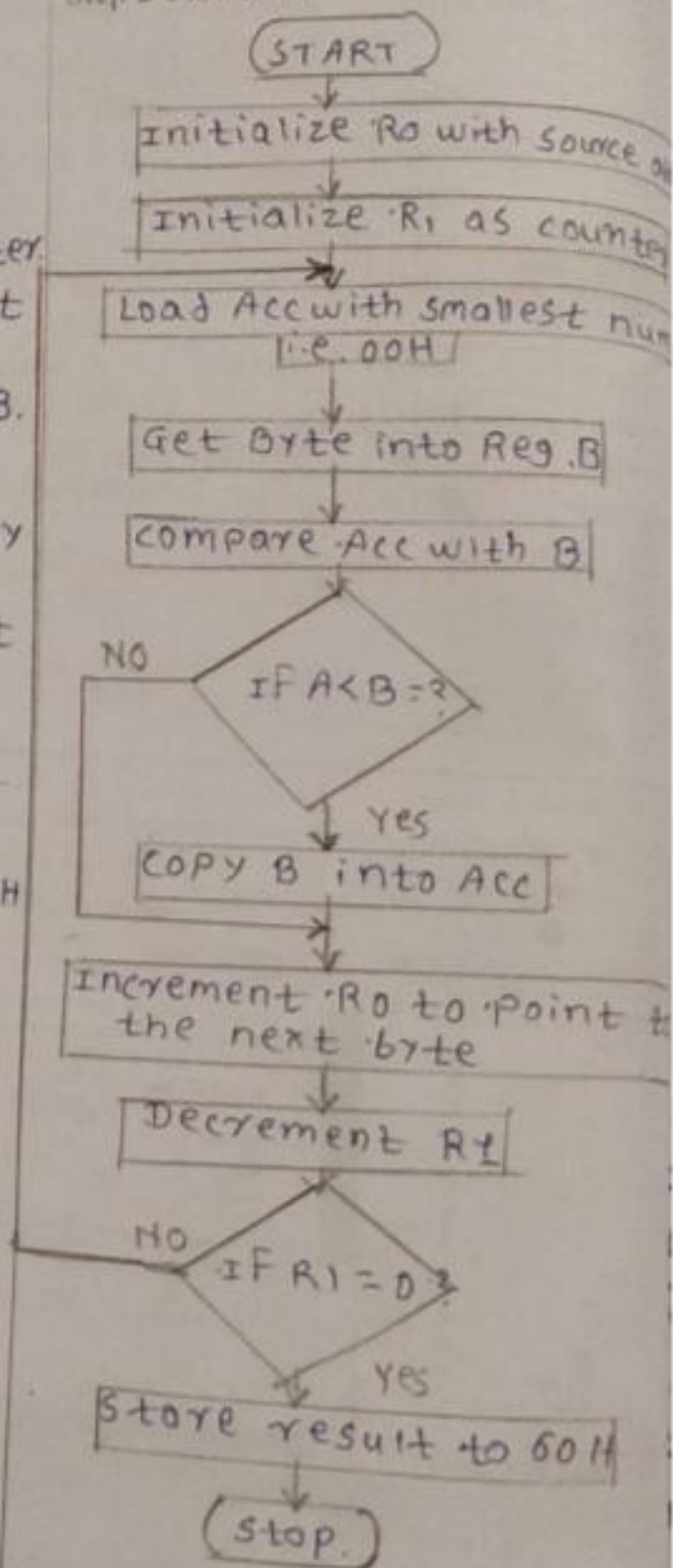
Both of these instructions are used to find the largest/smallest number.

Problem statement for student: Write ALP to find largest number from the given five stored in memory locations 20H onwards and store the result in location 60H

Step 1: Algorithm

- 1) Start.
- 2) Initialize R0 with source address.
- 3) Initialize R1 as counter.
- 4) Load Acc with smallest number i.e. 00H.
- 5) Get first number in B.
- 6) compare Acc with B.
- 7) If B is large then copy it into A.
- 8) Increment R0 to point to the next data.
- 9) Decrement R1 & if it is not zero then goto steps.
- 10) store the result at 60H present in Acc.
- 11) stop.

Step 2: Flowchart



Step 3: Assembly Language Program

Memory Address	Hex Code	Label	Mnemonic	Comments
			ORG 000H	
0000H	7905		MOV R1, #05H	R1 set as counter.
0002H	7820		MOV R0, #20H	Memory pointer.
0004H	7400		MOV A, #00H	Smallest number in A
0006H	8650	UP:	MOV B, @R0	Get first number in B
0008H	B5F000		CINE A, B, DOWN	compare A with B.
000BH	5002	DOWN:	JNC A-SMALL	if B is large then bring it into A
000DH	E5E0		MOV A, B	
000FH	08	A-SMALL	INC R0	Increment R0 to point to next data
0010H	D9F4		DJNZ R1, UP	Repeat till all data transfer.
0012H	F560		MOV 60H, A	Move largest no. in 60H location
			END	

XI Resources Used

Sr. No.	Name of Resource	Specification	Quantity
1	Desktop PC	loaded with open source IDE	1

XII Actual Procedure Followed (use blank sheet provided if space not sufficient)

- write program : (1) start keil by double clicking on keil icon.
- (2) create a new project (3) select device for target.
- (4) Double click on ATME1 or INTEL & select 8051AH or ATME1.
- (5) type the program in text editor & save as .asm or .a51, etc.

Precautions Followed (use blank sheet provided if space not sufficient)

1) check rules/syntax of assembly programming.

Observations for sample program (use blank sheet provided if space not sufficient)

Before execution		After execution	
Memory location	Data	Memory location	Data
40H	01H	50H	01H
41H	02H		
42H	03H		
43H	04H		
44H	05H		

XVI Interpretation of Results (Give meaning of the above obtained results)

1) we select summary location 20H, to 24H
& their respective data 01 to 05H.

XVII Conclusions and Recommendation

using instruction smallest & largest
no. are obtained instruction location.

XVIII Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design such questions so as to ensure the achievement of identified CO.

1. State the need of counter in finding the largest or smallest number.
2. List any four instructions to exchange data and give their syntax.
3. Write and execute an ALP to count zero's in a data 64H
4. Write and execute an ALP to find the average of five numbers

[Space for Answers]

1) → In order to find largest or smallest
no. memory no. are given in series
of Data. If the no. of data given series
are 10 then it is necessary to find
larger no. of the number by comparing
them to the find largest or smallest
it is necessary to have amount of
value 10. This count to have a
count taken in any of the register
R0 to RT.

~~2)~~ Instruction use for setting syntax
MOV R0, #count

2) → 1) MOV C destination - byte →, (source - byte)

MOV instruction copies the value of source byte into destination byte.

(ii) MOV DPTR, #data byte

MOV DPTR, #data 16 loads the data pointer with 16 bit data indicated instruction

3) → ORG 00H

LJMP MAIN

ORG 40H

MAIN :- MOV R0, #12

MOV R1, #DP

MOV A, R0

LOOP : RLCA

IN LABEL 1: A1

INC R2

SJMP LABEL 2

LABEL 1 :- INC R3

4) → MOV R0, #040H

MOV R1, #00AH

CLR A;

UP: ADD, @R0

INC R0

NEXT: DPTR, UP (C: DADS)

MOV 0x30A

MOV 8x31, R1

Practical No. 6: Write ALP for arranging numbers in ascending / descending order stored in external memory locations.

- I **Practical Significance**
Sorting is any process of arranging information systematically in ascending or descending order. This allows us to write better programs like indexing to fetch the information faster; allows faster search techniques, removes duplicate information and has many uses in statistical applications. This practical will help the students to develop skills to understand how to access data from external memory and use of branch instructions.
- II **Relevant Program Outcomes (POs)**
 - **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunications engineering related problems.
 - **Experiments and practice:** Plan to perform experiments and practices to use the results to solve broad-based Electronics and Telecommunication engineering problems.
 - **Engineering tools:** Apply relevant Electronics and Telecommunications technologies and tools with an understanding of the limitations.
- III **Competency and Practical Skills**
This practical is expected to develop the following skills for the industry identified competency 'Maintain microcontroller based systems':
 - Develop the code for the given problem.
 - Debug the code and rectify the errors.
 - Run the code
 - Use of application of instructions related to external memory.
- IV **Relevant Course Outcome(s)**
 - Interpret the program for 8051 in assembly language for the given operations.
- V **Practical Outcome**
 - Write ALP for arranging numbers in ascending /descending order stored in external memory locations
- VI **Relevant Affective domain related Outcome(s)**
 - Follow ethical practices.
- VII **Minimum Theoretical Background**
Ascending order/Descending order
The block of data consists of numbers in random order, to arrange these numbers in ascending or descending order bubble sort method is used.
If the given block of data has to be sorted in ascending order, then bubble sort will start by comparing the first element of the block with the second element, if the first element is greater than the second element, it will swap both the elements, and then move on to compare the second and the third element, and so on.
Byte counter –to access data from block of data
Pass counter –to repeat this comparison are required to arrange the numbers in ascending or descending order.

1) Problem statement for student: Write ALP stored in external memory location 2000H.

Step 1-Algorithm

- 1) start.
- 2) Initialize the register & data pointer.
- 3) Get first two elements in registers A & B.
- 4) compare the two element of data. If value of B register is low then exchange A & B data else increment pointer & decrement register R3.
- 5) Increment DPTR to point to the next element.
- 6) Repeat the process for all numbers.
- 7) stop.

Address	Hex	Assembly	Comments
0000H	7005	MOV R0, #00H	Counter
0001H	18	MOV R0, #00H	Counter-1
0002H	1800	MOV R0, #00H	Counter-2
0003H	300000	MOV R0, #00H	Counter-3
0004H	ED	MOV R0, #00H	Counter-4
0005H	F5F0	MOV R0, #00H	Counter-5
0006H	F5	MOV R0, #00H	Counter-6
0007H	ED	MOV R0, #00H	Counter-7
0008H	85F002	MOV R0, #00H	Counter-8
0009H	5008	MOV R0, #00H	Counter-9
000AH	4008	MOV R0, #00H	Counter-10
000BH	A5F0	MOV R0, #00H	Counter-11
000CH	CB	MOV R0, #00H	Counter-12
000EH	F0	MOV R0, #00H	Counter-13
000FH	1582	MOV R0, #00H	Counter-14
0010H	F8	MOV R0, #00H	Counter-15
0011H	F0	MOV R0, #00H	Counter-16
0012H		MOV R0, #00H	Counter-17
0013H		MOV R0, #00H	Counter-18
0014H	A3	MOV R0, #00H	Counter-19
0015H	D9E9	MOV R0, #00H	Counter-20
0016H	D9E2	MOV R0, #00H	Counter-21
0017H	80FE	MOV R0, #00H	Counter-22
0018H		MOV R0, #00H	Counter-23
0019H		MOV R0, #00H	Counter-24
001AH		MOV R0, #00H	Counter-25
001BH		MOV R0, #00H	Counter-26
001CH		MOV R0, #00H	Counter-27
001DH		MOV R0, #00H	Counter-28
001EH		MOV R0, #00H	Counter-29
001FH		MOV R0, #00H	Counter-30

XI Resources Used

S. No.	Instrument /Components	Specification	Quantity
1.	Desktop PC	loaded with open source IDE	1.

XII Actual Procedure Followed (use blank sheet provided if space not sufficient)

- Write program: (1) Start keil by double clicking on keil icon. (2) create a new project. (3) select device for target (4) Double click on ATMEL or INTEL & select 80C51AH or AT89C51. (5) Type the program in text editor & save as .asm or .a51.
- compile the program: (6) Right click on source group & build the target. (7) check for any errors in the oip window & remove if any.
- Run, Debug the program: (8) click on debug & start simulation & start/stop debug session. (9) Run the program step by step (10) observe the output on the project window. It will display internal registers of 8051 & their contents. (11) Note down the values of external memory location 3000H in the observation table.

XIII Precautions Followed (use blank sheet provided if space not sufficient)

1. check rules / syntax of assembly programming.

XIV Observations for sample program (use blank sheet provided if space not sufficient)

Before execution		After execution	
Memory location	Data	Memory location	Data
3000H		3000H	
3001H		3001H	
3002H		3002H	
3003H		3003H	
3004H		3004H	
3005H		3005H	

XV Results (Output of the Program)

.....

.....

.....

.....

Microcontroller Application (16)

XVI Interpretation of Results (Give meaning of the above obtained results)

written ALP for arranging numbers in
Ascending & Descending order stored in external
memory location.

XVII Conclusions and Recommendation (Actions/decisions to be taken based on
interpretation of results).

By performing this practical we conclude
how to develop code for arranging the
data in external memory in proper way.

XVIII Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design
such questions so as to ensure the achievement of identified CO

1. Write a program to (a) load the accumulator with the value 55H, and (b) complement
ACC 7 times.
2. Discuss the program ROM space allocation for each of the following cases. (a) $EA^+ = V_{CC}$
for the 8751 (89C51) chip. (b) $EA^+ = V_{CC}$ with both on-chip and off-chip ROM for
the 8751. (c) $EA^+ = V_{CC}$ with both on-chip and off-chip ROM for the 8752.

[Space for Answers]

- 1) what is competency & practical skill required
for performing above practical?
- 2) what is use of byte counter?

1) → Develop the code for given problem.
Debug the code & rectify the error.
Run the code.
use of application of instructions
related to external memory.

XVI Interpretation of Results (Give meaning of the above obtained results)
 written ALP for arranging numbers in
 Ascending & Descending order stored in external
 memory location.

XVII Conclusions and Recommendation (Actions/decisions to be taken based on
 interpretation of results).

By performing this practical we conclude
 how to develop code for arranging the
 data in external memory in proper way.

XVIII Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design
 such questions so as to ensure the achievement of identified CO

1. Write a program to (a) load the accumulator with the value 55H, and (b) complement
 ACC 7 times.
2. Discuss the program ROM space allocation for each of the following cases. (a) EA⁺
 for the 8751 (89C51) chip. (b) EA⁺ = Vcc with both on-chip and off-chip ROM for
 the 8751. (c) EA⁺ = Vcc with both on-chip and off-chip ROM for the 8752.

[Space for Answers]

- 1) what is competency & practical skill required
 for performing above practical?
- 2) what is use of byte counter?

1) → Develop the code for given problem.
 Debug the code & rectify the error.
 Run the code.
 Use of application of instructions
 related to external memory.

2) → Byte counter used for access data from block of data.

Practical No. 7: Write an ALP to generate delay using register.

I Practical Significance

Creating a delay is most important and widely used function in almost all microcontroller-based programming. Its use varies from creating a pause to allow microcontroller to interact with real world, like blinking of LED, monitoring a switch or input at regular interval. In signal processing, a "delay" is used to monitor signal at a predetermined sampling rate. This practical will help the students to develop skills to create the time delay by using instruction machine cycles and microcontroller crystal frequency.

II Relevant Program Outcomes (POs)

- **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunications engineering related problems.
- **Experiments and practice:** Plan to perform experiments and practices to use the results to solve broad-based Electronics and Telecommunication engineering problems.
- **Engineering tools:** Apply relevant Electronics and Telecommunications technologies and tools with an understanding of the limitations.

III Competency and Practical Skills

This practical is expected to develop the following skills for the industry identified competency 'Maintain microcontroller based systems':

- Develop the code for given problem.
- Debug the code and rectify the errors.
- Run the code
- To generate delays by appropriate delay calculations.

IV Relevant Course Outcome(s)

- Interpret the program by using timer, interrupt and serial ports/parallel ports.

V Practical Outcome

- Write an ALP to generate delay using register.

VI Relevant Affective domain related Outcome(s)

- Follow ethical practices.

VII Minimum Theoretical Background

The delay length in 8051 microcontroller depends on two factors:

1. The crystal frequency.
2. The number of clock per machine.

Machine cycle for the 8051

The CPU takes a certain number of clock cycles to execute an instruction. In the 8051 family, these clock cycles are referred to as *machine cycles*. In the 8051 family, the length of the machine cycle depends on the frequency of the crystal oscillator connected to the 8051 systems. The crystal oscillator, along with on-chip circuitry, provides the clock source for the 8051 CPU. The frequency of the crystal connected to the 8051 family can vary from 4 MHz to 30 MHz, depending on the chip rating and manufacturer.

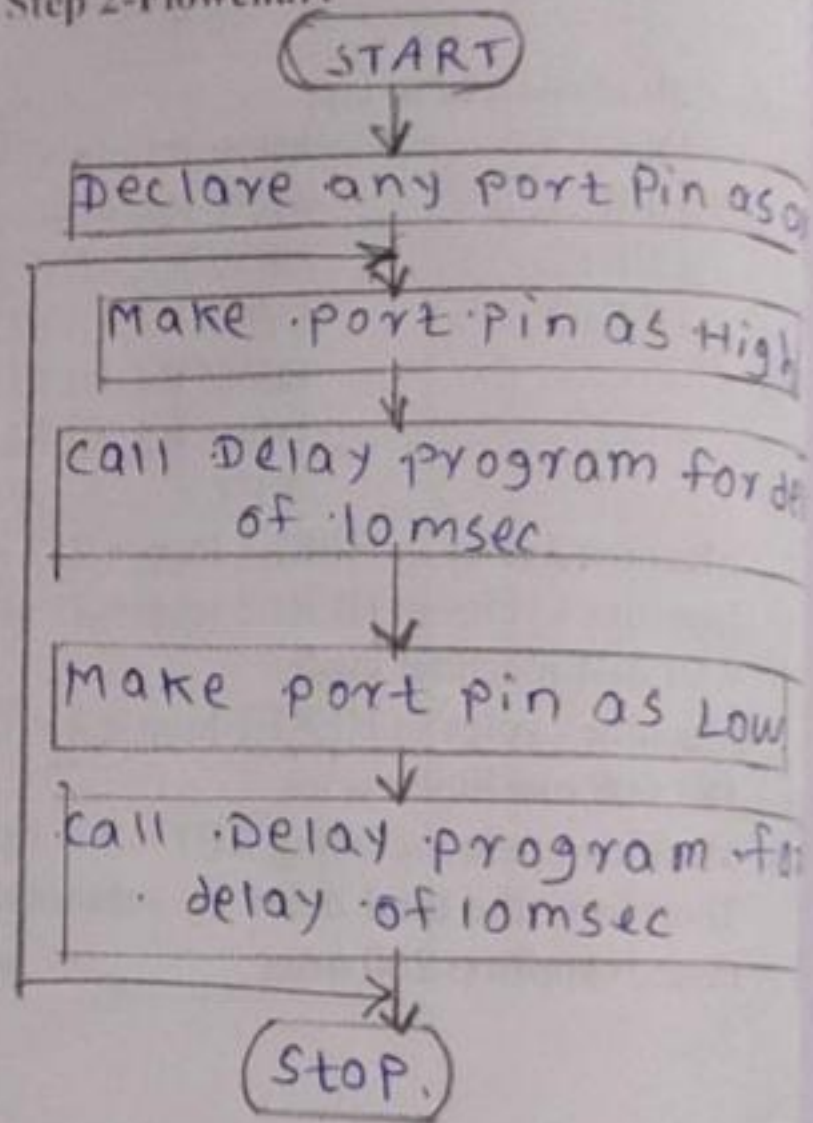
Fig 7.2 Output Window

1. Problem statement for student: Write a subroutine to generate a delay of 10ms

Step 1-Algorithm

- 1) Start
- 2) Declare any port pins as output.
- 3) Make port pin as High.
- 4) call Delay program for delay of 10 msec.
- 5) make port pin as low.
- 6) call Delay program for delay of 10 msec.
- 7) Go back to step 3 to Repeat.
- 8) Stop.

Step 2-Flowchart



Step 2: Assembly Language Program

Memory Address	Hex Code	Label	Mnemonics	Comments
			ORG 0000H	
0000H	C292		CLR P1.2	Declare P1.2 as output
0002H	D292	START:	SETB P1.2	Make P1.2 High
0004H	110C		CALL DELAY	call delay of 10msec
0006H	C292		CLR P1.2	Make P1.2 Low
0008H	110C		CALL DELAY	call delay of 10msec
000AH	80F6		SJMP START	Repeat Forever
000CH	7A12	DELAY:	MOV R2, #18	Delay Program
000EH	7BFF		MOV R3, #255	
0010H	DBFE		DJNZ R3, UP	
0012H	DAFA		DJNZ R2, UP1	
0014H	22		RET	Return to main program.
			END	

XI Resources Used

S. No.	Instrument /Components	Specification	Quantity
1.	Desktop PC	loaded with open source IDE	1

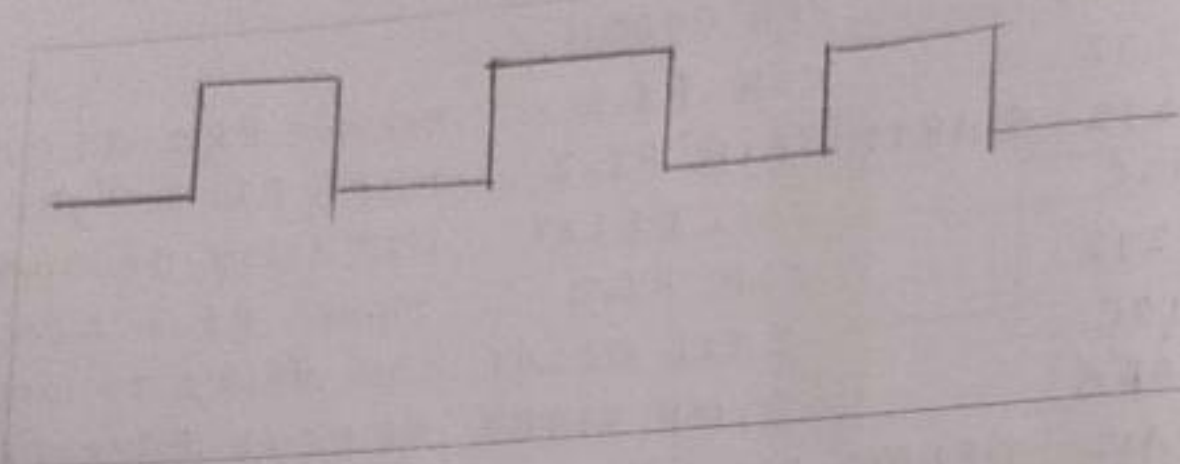
XII Actual Procedure Followed (use blank sheet provided if space not sufficient)

- Write program: (1) Start Keil by double clicking on keil icon. (2) create a new project. (3) Select device for target (4) Double click on ATMEL or INTEL & select 80C51AH or AT89C51. (5) Type the program in the text editor & save as .asm or .a51.
- Compile the program: (6) Right click on source group & build the target (7) check for any errors in the o/p window & remove if any.
- Run, Debug the program: (8) click on debug & start simulation & start/stop debug session. (9) Run the program step by step (10) observe the output on the project window. It will display all internal registers of 8051 & their contents. (11) observe the waveform on logic analyser & sketch the same in observation.

XIII Precautions Followed (use blank sheet provided if space not sufficient)

1. check rules/syntax of assembly programming.

XIV Observations for sample program (use blank sheet provided if space not sufficient)
Draw the waveform and mention the delay



XV Results (Output of the Program)

P_1 is tagged with delay of 200ms.

XVI Interpretation of Results (Give meaning of the above obtained results)

Square waves of any frequency can be generated using 8051 IC.

XVII Conclusions and Recommendation

By performing this practical we know that how to generate address.

XVIII Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design such questions so as to ensure the achievement of identified CO

1. For an 8051 microcontroller how long will it take to execute the following instruction. The crystal frequency is 11.0592 MHz i) MOV R3, 55H ii) DEC R2
2. On what factors does delay length depend upon?
3. For three different 8051 versions, what will be the machine cycle for the following crystal frequencies:
i) 11.0592 MHz ii) 16 MHz iii) 20 MHz Find period of machine cycle in each case

[Space for Answers]

1) $\Rightarrow$ The machine cycle for a system of 11.0592 MHz is 1.085 μ s.

2) → Delay length depend on following two function.

1) crystal function.

2) the no. of clock per machine.

3) → i) $11.592 \text{ MHz} / 12 = 921.16 \text{ KHz}$.

Machine cycle is $\frac{1}{921.16} \text{ KHz} = 11.085 \mu\text{s}$.

ii) $16 \text{ MHz} = 1.333 \text{ MHz}$

machine cycle (MC) = $\frac{1}{33} \text{ MHz} = 0.75 \mu\text{s}$.

XIX References / Suggestions for further reading

1. The 8051 Microcontroller and Embedded system Using Assembly and C - Muhammad Ali Mazidi, Janice Gillispie Mazidi, Rolin D. McKinlay - Pearson Prentice Hall, 2nd edition, Delhi, 2008, ISBN 978-8177589030
2. <https://www.youtube.com/watch?v=TWMAi3oimM>

XX Assessment Scheme

Performance Indicators		Weightage
Process related: 15 Marks		60%
1	Use of IDE tools for programming	20%
2	Coding and Debugging ability	30%
3	Follow ethical practices.	10%
Product related: 10 Marks		40%
4	Correctness of algorithm/ Flow chart	20%
5	Relevance of output of the problem definition	15%
6	Timely Submission of report, Answer to sample questions	05%
Total (Marks)		100 %

Name of Team Members

- 1
- 2
- 3
- 4

Marks Obtained			Dated signature of Teacher
Process Related (15)	Product Related (10)	Total (25)	

Practical No. 17: Write an ALP to transfer 8-bit data serially on serial port

- I. Practical Significance**
Many applications require microcontrollers to either accept the data in serial form or output the data in serial form. Serial communication is commonly used in applications such as industrial automation systems, scientific analysis and certain consumer products. This practical will help the students to develop skills to understand the concepts of serial port and how they are interfaced with microcontroller.

II. Relevant Program Outcomes (POs)

- **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunications engineering related problems.
- **Experiments and practice:** Plan to perform experiments and practices to use the results to solve broad-based Electronics and Telecommunication engineering problems.
- **Engineering tools:** Apply relevant Electronics and Telecommunications technologies and tools with an understanding of the limitations.

III. Competency and Practical skills

This practical is expected to develop the following skills for the industry identified competency 'Maintain microcontroller based systems':

- Develop the code for the given problem
- Debug the code and rectify the errors.
- Run the code
- Select appropriate value of TH1 register needed for the corresponding baud rate

IV. Relevant Course Outcome(s)

- Interpret the program by using timer, interrupt and serial ports /parallel ports

V. Practical Outcome(PROs)

- Write an ALP to transfer 8 bit data serially on serial port

VI. Relevant Affective domain related Outcome(s)

- Follow ethical practices.

VII. Minimum Theoretical Background

In serial communication two formats of data transfer are used 1) Asynchronous 2) Synchronous.

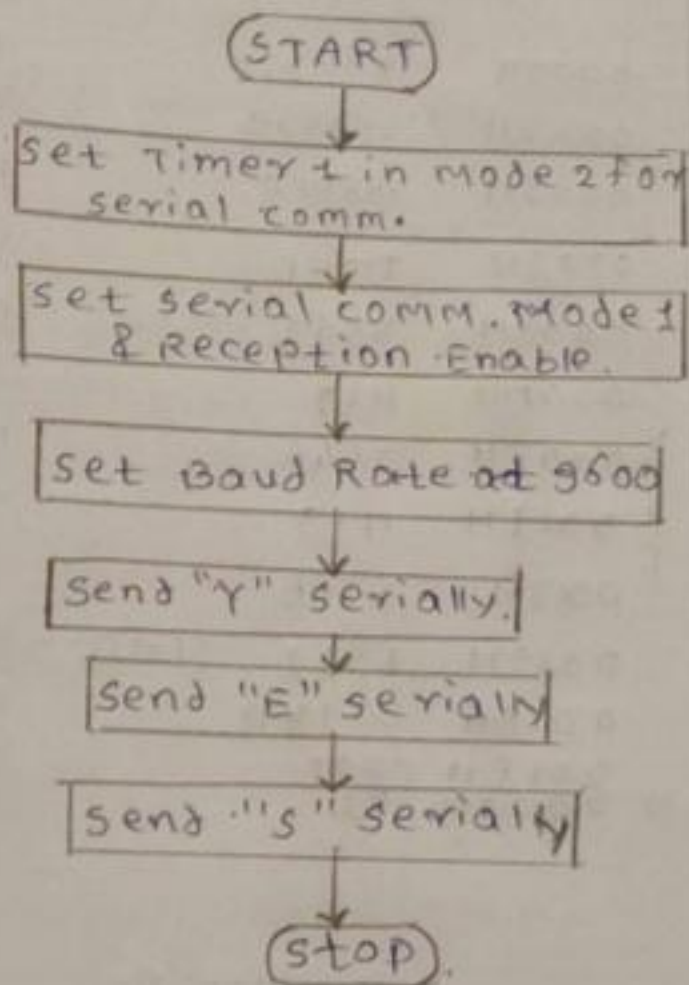
One of the major differences is that in Synchronous Transmission, the sender and receiver should have synchronized clocks before data transmission. Whereas Asynchronous Transmission does not require a clock, but it adds a parity bit to the data before transmission.

Problem statement 1 for students: Write a program to transfer to send letter "YES" at baud rate 9600 continuously.

Step 1-Algorithm

- 1) start.
- 2) set timer 1 in mode 2 for serial comm.
- 3) set serial comm. mode 1 & Reception Enable.
- 4) set Baud rate at 9600.
- 5) send "Y" serially.
- 6) send "E" serially.
- 7) send "S" serially.
- 8) stop.

Step 2-Flowchart



Step 3: Assembly Language Program				Comments
Memory Address	Hex Code	Label	Mnemonics	
			ORG 0000H	
0000H	758320		MOV TMOD, #20H	Set timer 1 in mode 2
0003H	759850		MOV SCON, #50H	Set serial comm. mode 1 Reception enable
0006H	758DFD		MOV TH1, #0FDH	Set Baud rate at 9600
0009H	D28E		SETB TR1	Start Timer 1.
000BH	7459		MOV A, #'Y'	Copy 'Y' into Accumulator
000DH	1119		ACALL SEND	Send 'Y' serially.
000FH	7445		MOV A, #'S'	Copy 'S' into Accumulator
0011H	1119		ACALL SEND	Send 'S' serially.
0013H	80FE		SJMP \$	Stay Here.
0019H	F599	SEND:	MOV SBUF, A	Load char. to be sent to the SBUF register
001BH	3099FD		JNB TI, HR	Wait till char. transfer & TI become set.
001EH	C299		CLR TI	Clear TI for next transfer
XI 0020H	124		RET	Return to main program
			END	Stop.

Sr. No.	Instrument /Components	Specification	Quantity
1.	Dekstop PC	loaded with open source IDE	1
2.	Interfacing circuit	Microcontroller kit	1.

XII Actual Procedure Followed (use blank sheet provided if space not sufficient)

- Write program: (1) start keil by double clicking on keil (2) create a new project. (3) select device for target. (4) Double click on ARMEL or INTEL & select 80C51AH or AT89C51 (5) Type the program in text editor & save as .asm
- compile the program: (6) Right click on source group & build the target. (7) check for any errors in the D/P window & remove if any.
- Run, Debug the program: (8) click on debug & start simulation & start/stop debug session. (9) Run the program step by step. (10) observe the o/p on the project window. The output can be observed in UART 1 window.
- (11) Hyper terminal, a windows XP application, can be used to receive & transmit serial data through RS232. To open hyper terminal, go to start menu & select

all programs. go to Assesories, click on communications & select hyper terminal.

2) To start a new connection, go to file menu & click on new connection. The connection window opens up. Give a name to your connection & select 1st icon & click on ok. connection property window opens here. select bit rate as 9600 bps. Data bits 8, parity as none, stop bit flow control none. & click ok. Now the serial data can be read on hyper terminal.

3) In program, Timer 1 is used with auto reload setting. the baud rate is fixed to 9600 bps by loading TH1 to 0xFD. The value 0x50 is loaded in the SCON register. This will initialize the serial port in mode 1. The program continuously receives a character (say 'a') from the serial port of computer & transmits it back.

XIII Precautions Followed (use blank sheet provided if space not sufficient)

check rules/syntax of assembly programming.

XIV Observations for sample program (use blank sheet provided if space not sufficient)
(UART window)

XV Result (Output of the Program)

XVI Interpretation of Results (Give meaning of the above obtained results)

By sending 3 alphabets (Y, E, S) serially at Baud rate 9600 we completed our practical.

XVII Conclusions and Recommendation

Written a program to transfer/send letter 'YES' at Baud rate 9600 continuously.

XIII Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design more such questions so as to ensure the achievement of identified CO.

1. State the SFR used to transfer character by serial communication.
2. With XTAL = 11.0592 MHz, find the TH1 value needed to have the following baud rates. (a) 9600 (b) 2400 (c) 1200
3. Name the timer of 8051 used to set the baud rate.

[Space for Answers]

1) → ACC, B, DPH, DPL, IE, IP, P0, P1, P2, P3, PCON, PSW, SCON, SBUF, SP, TMOD, TCON, TLO, TH0, TL1, TH1 are the SFR used to transfer character.

2) → For the 8051 the "Timer1" is used to generate the baud rate in Auto reload mode.

Practical No. 9: Interface LED with microcontroller and turn it ON with microcontroller interrupt.

- I **Practical Significance**
LED is most common semiconductor device used in many electronic system as visual indicator or signal transmission / power indication purposes. The LEDs are also used in design message display boards and traffic control signal lights etc. Interrupts are an important feature of Microcontroller. This practical will help the students to develop skills to understand the fundamental interfacing concept for 8051 microcontrollers and significance of Interrupts.

- II **Relevant Program Outcomes (POs)**
- **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunications engineering related problems.
 - **Experiments and practice:** Plan to perform experiments and practices to use the results to solve broad-based Electronics and Telecommunication engineering problems.
 - **Engineering tools:** Apply relevant Electronics and Telecommunication technologies and tools with an understanding of the limitations.
 - **Lifelong learning:** Engage in independent and life-long learning activities in the context of technological changes also in the Electronics and Telecommunication engineering and allied industry

- III **Competency and Practical Skills**
This practical is expected to develop the following skills for the industry identified competency 'Maintain microcontroller based systems':

- Interface LED and switch to microcontroller.
- Use of key to provide external interrupt on pin P3.2.

- IV **Relevant Course Outcome(s)**
- Interpret the program by using timer, interrupt and serial ports /parallel ports.

- V **Practical Outcome**
- Interface LED with microcontroller and turn it ON with microcontroller interrupt.

- VI **Relevant Affective domain related Outcome(s)**
- Follow safe practices.
 - Maintain tools and equipment.
 - Follow ethical practices.

VII **Minimum Theoretical Background**

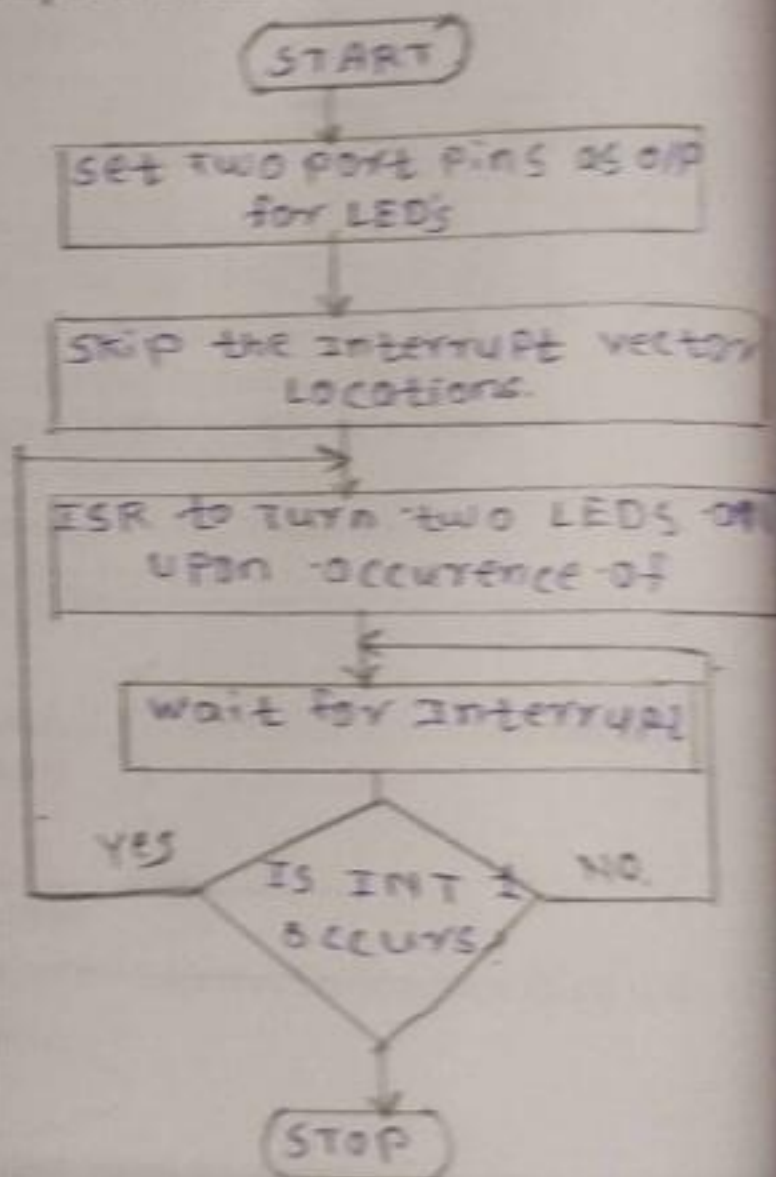
As its name suggests, **Interrupt** is a subroutine call that interrupts microcontrollers main operations or work and causes it to execute any other program, which is more important at the time of operation. The feature of Interrupt is very useful as it helps in case of emergency operations. An Interrupts gives us a mechanism to put on hold the ongoing operations, execute a subroutine and then again resume its original task.

Generally five interrupt sources are there in 8051 Microcontroller. Out of these, **INT0** and **INT1** are external interrupts that could be negative edge triggered or low level triggered. They are located on pins P3.2 and P3.3 of port 3 respectively. They are enabled or

Step 1: Algorithm

- 1) start
- 2) Set two port pins as output for LEDs.
- 3) Skip the interrupt vector locations.
- 4) ISR to turn two LEDs on upon occurrence of INT1.
- 5) Enable interrupt INT1.
- 6) wait for interrupt.
- 7) If INT0 occurs then go to step 4.
- 8) stop.

Step 2: Flowchart



Step 3- Assembly Language Program

Memory Address	Hex Code	Label	Mnemonics	Comments
			ORG 0000H	
0000H	C293		CLR P1.3	Set port pins as output.
0002H	C294		CLR P1.4	
0004H	020030		LJMP MAIN	
			ORG 0013H	Skip the interrupt vector location
0013H	D293		SETB P1.3	Interrupt service Routine.
0015H	D294		SETB P1.4	Turn ON LED 1
0017H	7A0F		MOV R2, #15	Turn ON LED 2
0019H	7BFF	UP3:	MOV R3, #255	Delay program.
001BH	7CFF	UP2:	MOV R4, #255	
001DH	DCFE	UP1:	DJNZ R4, UP1	
001FH	DBFA		DJNZ R3, UP2	
0021H	DAF6		DJNZ R2, UP3	
0023H	C293		CLR P1.3	Turn off LED 1
0025H	C294		CLR P1.4	Turn off LED 2
0027H	32		RETI	Return to Main program
			ORG 0030H	Main program.
0030H	75A884	MAIN:	MOV IE, #10000100B	Enable interrupt INT1.
0033H	80FE	HERE:	HERE: SJMP HERE	Wait for interrupt
			END	Stop.

XII Resources Used

S. No.	Instrument /Components	Specification	Quantity
1.	Desktop PC	Loaded with open source IDE	1.
2.	Interfacing circuit	Microcontroller kit	1

XIII Actual Procedure Followed

- (1) write algorithm for given problem. (2) Draw flowchart for the same. (3) Develop assembly program using Integrated Development environment (Keil IDE) or any other relevant software tool. (4) Debug program on IDE. (5) Execute program on IDE. (6) Create hex file for the program. (7) Download hex code in EPROM/Flash memory of microcontroller. (8) Interface LED to microcontroller as per circuit diagram shown in Fig. (9) observe the LED to glow when external interrupt occurs on P3.2.

XIV Precautions Followed (use blank sheet provided if space not sufficient)

1. Use always current limiting resistor before LED connected to microcontroller.

XV Observations (use blank sheet provided if space not sufficient)

LED will become ON (ON/OFF) after occurrence of INT1. (INT0/INT1)
Interrupt

XVI Result (Output of the Program)

using microcontroller interrupts LED will become ON.

XVII Interpretation of Results (Give meaning of the above obtained results)

Microcontroller interrupt INT1 occurs the after LED was glow.

XVIII Conclusions and Recommendation

By performing this practical we interface two LED's with microcontroller & turn them ON with microcontroller interrupts.

Note: Below given are few sample questions for reference. Teacher must design more such questions so as to ensure the achievement of identified CO

1. With a single instruction, show how to disable all the interrupts.
2. List the interrupts of 8051 with vector address, port pin, flag affecting, priority level.

[Space for Answers]

1) → use the "#pragma disable" directive to disable interrupts for an entire function. Then you can create a function that sets up the ISR or whatever & not even DEAL with the EA = 0 bit.
for example.

```
#pragma disable
void ISR_setup(void)
{
    // set timer reload
    // set timer mode
    // Etcetera
}
```

2) → 8051 has 5 interrupt signals i.e. INTO, TFO, INT1, IE, & RI/TI. Each interrupt can be enabled or disabled by setting bits of the IE register & the whole interrupt system can be disabled by clearing the EA bit of the same register.

Practical No. 10: Develop an ALP to generate pulse and square wave by using Timer delay.

Practical Significance

In 8051 timers are used to generate delays or as counters to count events happening outside the microcontroller. In time required applications two available 16 bit timers are operated in different modes to generate specific delay. This practical will help the students to develop skills to program timers and generate delays.

Relevant Program Outcomes (POs)

- **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunications engineering related problems.
- **Experiments and practice:** Plan to perform experiments and practices to use the results to solve broad-based Electronics and Telecommunication engineering problems.
- **Engineering tools:** Apply relevant Electronics and Telecommunications technologies and tools with an understanding of the limitations.

Competency and Practical Skills

This practical is expected to develop the following skills for the industry identified competency 'Maintain microcontroller based systems':

- Select appropriate mode of timer for application.
- Program two 16 bit timers/counters independently.
- Generate required time delay.

Relevant Course Outcome(s)

- Interpret the program by using timer, interrupt and serial ports /parallel ports.

Practical Outcome

- Develop an ALP to generate pulse and square wave by using Timer delay.

Relevant Affective domain related Outcome(s)

- Practice good housekeeping
- Demonstrate working as a leader/a team member.
- Follow ethical practices.

Minimum Theoretical Background

Four special function registers are connected with TIMER/COUNTER operation.

TMOD - Timer Mode Register

TCON - Timer Control Register

TH0, TL0 - Timer/Counter - 0

TH1, TL1 - Timer/Counter - 1

8085 Assembly Language Program

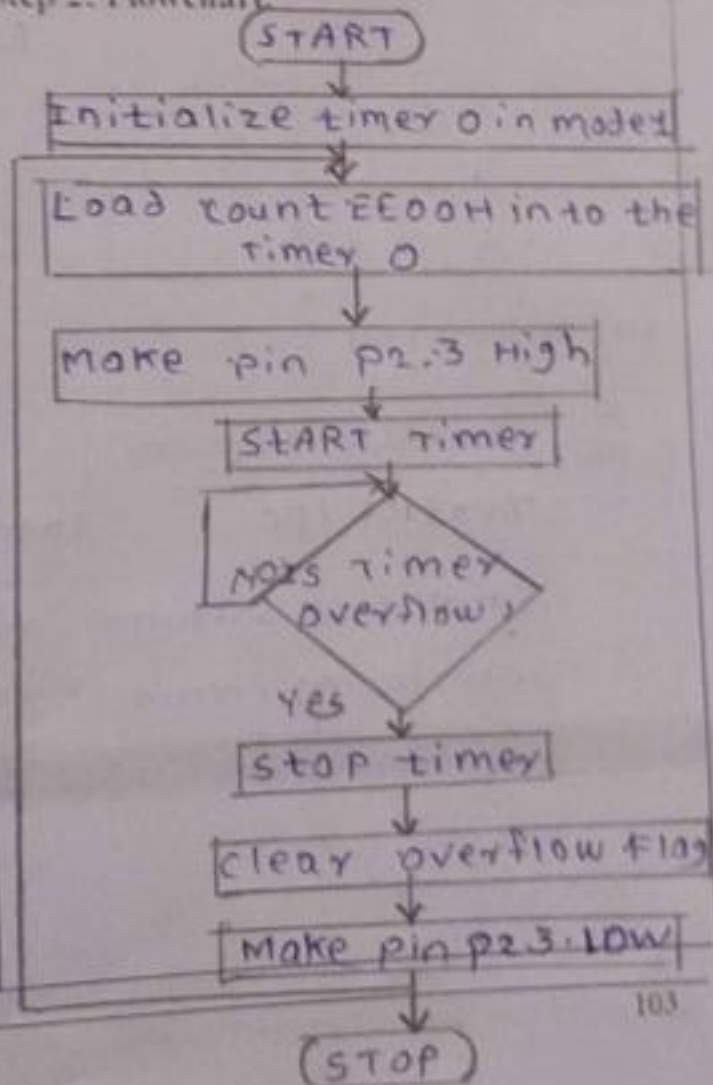
Memory Address	Hex Code	Label	Mnemonic	Comments
C0x0000	758904		ORG 0000H	
C0x0003	75AF2	HERE	MOV TMR0, 00H	Timer 0 mode 1
C0x0006	75CFF		MOV TH0, 0FH	TH0=0FH
C0x0009	B285		MOV TL0, 0FH	TL0=0FH
C0x000B	110F		CPL P1.5	toggle P1.5
C0x000D	80F4		ACALL DELAY	
C0x000F	D28C	DELAY:	SETB TR0	load TH, TL again
C0x0011	308DFD	AGAIN:	JNB TF0, AGAIN	start Timer 0
C0x0014	C28C		CLR TR0	monitor Timer 0 overflow flag until it rolls over
C0x0016	C28D		CLR TF0	stop Timer 0
C0x0018	22		RET	clear Timer 0 flag
			END	

Problem statement for student: Develop assembly program to generate pulse width of 5ms on P2.3 using timer 0 mode 1. Assume XTAL = 11.0592MHz

Step 1: Algorithm

- 1) start.
- 2) Initialize timer 0 in mode 1.
- 3) Load count EE00H into the timer 0.
- 4) Make pin P2.3 high.
- 5) Start timer.
- 6) wait for timer to be overflow.
- 7) stop timer.
- 8) clear overflow flag for next cycle.
- 9) make pin P2.3 low.
- 10) Go to step 3 for repeat.
- 11) stop.

Step 2: Flowchart



Step 3- Assembly Language Program

Memory Address	Hex Code	Label	Mnemonics	Comments
			ORG 0000H	
0000H	758701		MOV TMOD, #01H	Initialize timer 0
0003H	758CEE		MOV TH0, #0EEH	Load count EE00H into TH0
0006H	758A00	START:	MOV TLO, #00H	
0009H	D2A3		SETB P2.3	Make pin P2.3 high
000BH	D28C		SETB TRO	Start Timer 0
000DH	308DEF	ABC:	JNB TFO, ABC	wait for timer to be 0
0010H	C28C		CLR TRO	Stop timer
0012H	C28D		CLR TFO	clear overflow flag for timer 0
0014H	C2A3		CLR P2.3	Make pin P2.3 Low
0016H	80EB		SJMP START	Repeat Forever
			END	STOP

XII Resources Used

Sr. No.	Instrument /Components	Specification	Quantity
1.	Desktop /pc	latest processor	1
2.	Simulation software	Keil	1
3.	Interfacing circuits	microcontroller kit & CRO	1 Each.

VIII Actual Procedure Followed (use blank sheet provided if space not sufficient)

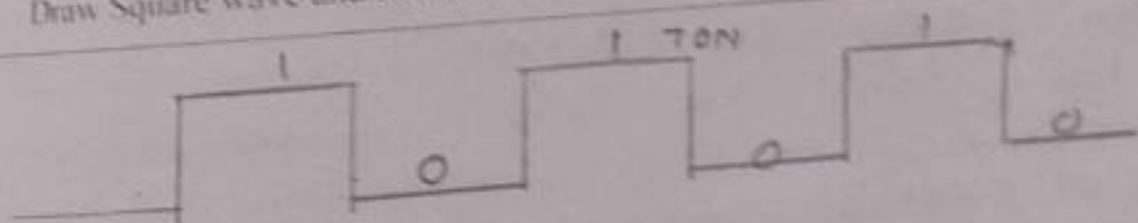
1. Write algorithm of given problem.
2. Draw flowchart for the same.
3. Develop assembly program using Integrated Development Environment (Keil IDE) or any other relevant software tool.
4. Debug program on IDE.
5. Execute program on IDE.
6. Create hex file for the program.
7. Download hex code in EPROM/Flash memory of microcontroller.
8. Connect CRO probe to port pin & observe waveform.
9. Measure ON time & off time on CRO & draw the same in observation table.

IX Precautions Followed (use blank sheet provided if space not sufficient)

1. Check rules/syntax of assembly programming.

X Observations for sample program (use blank sheet provided if space not sufficient)

Draw Square wave and show TON and TOFF time



XVI Result (Output of the Program)

Timer delay For $T_{ON} = T_{OFF} = 31.58 \mu s$
pulse width per value $F_{Hz} = 69.43 Hz$
Frequency can be at $31.222.5 Hz$

XVII Interpretation of Results (Give meaning of the above obtained results)

The square waveform drawn using
Ton & Toff time with delay off & pulse

By using timer we can generate delay of any frequency for square wave.

XIX Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design more such questions so as to ensure the achievement of identified CO

1. Calculate the number to be loaded into TH, TL to generate square wave of ON time = 3ms and OFF time = 10ms use mode 1. Assume XTAL of 22MHz.
2. To get a 2 ms delay, what number should be loaded into TH, TL using mode 1. Assume XTAL = 11.0592MHz.

[Space for Answers]

1) $\rightarrow$ delay time = 2ms

$$2\text{ms} = (65536 \cdot \text{initialize value} \times 1.085$$

$$1843 - (65536 \cdot \text{initial value})$$

$$(63692)_{10} = \text{f8ed}$$

$$\boxed{\text{TH} = 0F8H} \quad \text{TL} = 0CDH$$

2) $\rightarrow$ $T_{ON} = 2\text{ms}$, $T_{f2} = 1.5\text{ms}$

$$T_{OFF} = 10\text{ms}, T_{f2} = 5\text{ms}$$

for T_{ON} :

$$1.5\text{ms} (65536) - \text{initialize value } 5658H$$

$$(1627786)_{10} = \text{f542}$$

for T_{OFF} :

$$5\text{ms} = (65536 - \text{initial value} \times 545.4\text{ms})$$

$$9106 = (65536 - \text{initial})$$

$$(56570)_{10} = \text{dc32}$$

Practical No. 11: Interface 4 X 4 LED matrix with 8051 to display various pattern.

I

Practical Significance

LED matrix displays are used as stadium displays, decorative displays and as signals to human eye, to convey a message or meaning. LED matrix display interfaced with microcontroller I/O port to display characters and different patterns. This practical will help the students to develop skills to interface 4 X 4 LED matrix display with microcontroller and display various pattern.

II

Relevant Program Outcomes (POs)

- **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunications engineering related problems.
- **Experiments and practice:** Plan to perform experiments and practices to use results to solve broad-based Electronics and Telecommunication engineering problems.
- **Engineering tools:** Apply relevant Electronics and Telecommunication technologies and tools with an understanding of the limitations.

III

Competency and Practical Skills

This practical is expected to develop the following skills for the industry identified competency 'Maintain microcontroller based systems':

- Select appropriate type of LED grid for given application.
- Interface 4x4 LED matrix display to the microcontroller.
- Generate different patterns on LED matrix.

IV

Relevant Course Outcome(s).

- Interface the memory and I/O devices to 8051 microcontroller.

V

Practical Outcome

- Interface 4 X 4 LED matrix with 8051 to display various pattern.

VI

Relevant Affective domain related Outcome(s)

- Practice good housekeeping.
- Demonstrate working as a leader/a team member.
- Follow ethical practices.

VII

Minimum Theoretical Background

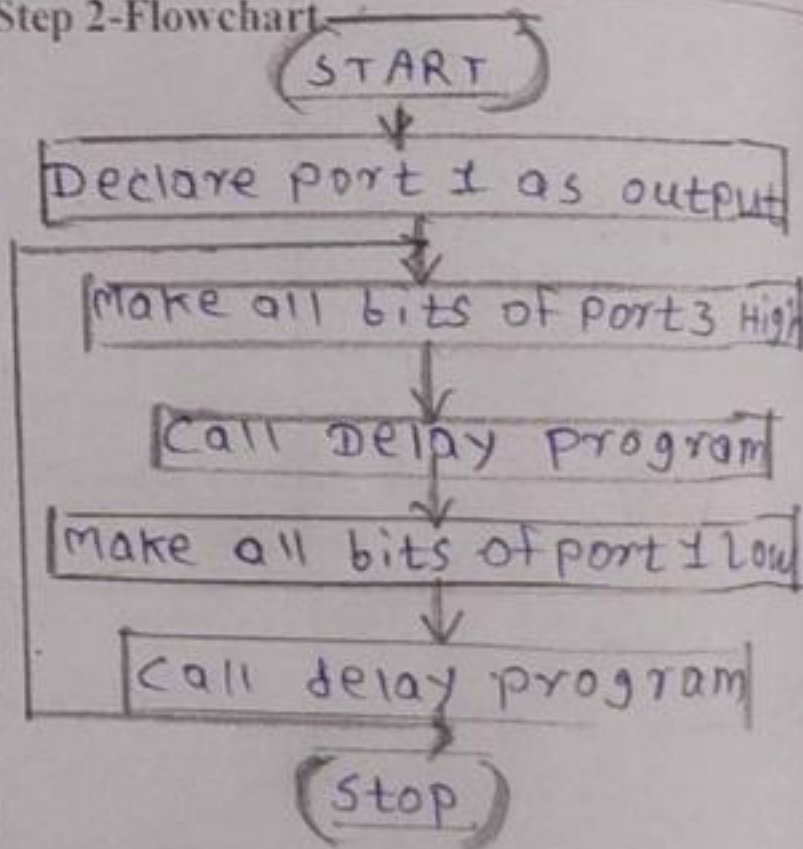
In LED dot matrix display the LEDs are connected at the column and row intersection of the matrix. LEDs in the same row are connected together and LEDs in the column are connected together. Transistors are act as switches and used to control LEDs in the matrix.

Problem statement for student: Develop assembly program to turn ON and OFF all LEDs connected to port 1 with some delay.

Step 1: Algorithm

- 1) start.
- 2) Declare port 1 as output port.
- 3) Make all bits of port 1 High.
- 4) call Delay program.
- 5) Make all bits of port 1 Low.
- 6) call Delay program.
- 7) Go back to step 3 to Repeat.
- 8) stop.

Step 2-Flowchart



Step 3- Assembly Language Program Code

Memory Address	Hex Code	Label	Mnemonics	Comments
0000H	759000		MOV P1, #00H	make port 1 as output port
0003H	7590FF	BACK:	MOV P1, #0FFH	Turn ON all LEDs
0006H	110F		ACALL DELAY	Wait for some time
0008H	759000		MOV P1, #00H	Turn OFF all LEDs
000DH	110F		SJMP BACK	Repeat Forever
000FH	8054	DELAY:	MOV R1, #0FFH	Delay program
0011H	79FF	UP1:	MOV R2, #0FFH	
0013H	7AFF	UP:	DJNZ R2, UP	
0015H	DAFE		DJNZ R1, UP1	
0017H	D9FA		RET	Return to main program
			END	

XII Resources Used

Sr. No.	Name of Resource.	Specification	Quantity
1.	computer / Laptop	Latest processor	1
2.	Simulation software	Keil	1
3.	Interfacing circuits	Microcontroller kit 4x4 LED MATRIX	1 each

XIII Actual Procedure Followed (use blank sheet provided if space not sufficient)

1. Write algorithm for given problem.
2. Draw flowchart for the same.
3. Develop assembly program using integrated development Environment (Keil IDE) or any other relevant software.
4. Debug program on IDE.
5. Simulate program on IDE.
6. Create hex file for the program.
7. Download hex code in EPROM / Flash memory of the Microcontroller.
8. Interface 4x4 LED matrix to 8051.
9. Observe different patterns on 4x4 LED matrix.

XIV Precautions Followed (use blank sheet provided if space not sufficient)

1) Always use driver circuit to interface LED matrix to microcontroller.

XV Observations for sample program (use blank sheet provided if space not sufficient)

HEX Value	LED status (ON/OFF)	
	Column	Row
F1H		
1FH		
11H		
88H		

XVI Results (Output of the Program)

XVII Interpretation of Results (Give meaning of the above obtained results)

we turned on an individual LED by setting its row & column pins to the proper logic 1.

XVIII Conclusions and Recommendation (Actions/decisions to be taken based on the interpretation of results).

By performing we learnt how to develop Assembly program to turn ON & OFF all LED's connected to port 1 with some delay.

XIX Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design more such questions so as to ensure the achievement of identified CO.

1. Give syntax of two instructions used in program to exchange data.
2. Write and execute a program to glow the second line display on 4x4 LED matrix display continuously.

[Space for Answers]

1) Give specifications of LED used in practical
→ specifications :- 1) current: 20 mA
2) Voltage drop: 1.9 to 2.1 V
3) power dissipation: 40 mW
4) Colour: Red.

2) what is operation of LED in this practical.

→ voltage $+5\text{ V}$ to anode with respect to cathode ground LED will turn ON.

Practical No. 12: Interface 7-segment display to display the decimal number from 0 to 9.

Practical Significance

In electronic displays like pricing menu at petrol pump, in metros, digital clocks and in many electronics appliances most commonly used display device is a 7-segment display (SSD). Seven segment displays are used to display decimal numbers from 0 to 9 and in some cases, basic characters also. This practical will help the students to develop skills to interface 7-segment display to microcontroller and display number from 0 to 9.

Relevant Program Outcomes (POs)

- **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunications engineering related problems.
- **Experiments and practice:** Plan to perform experiments and practices to use the results to solve broad-based Electronics and Telecommunication engineering problems.
- **Engineering tools:** Apply relevant Electronics and Telecommunications technologies and tools with an understanding of the limitations.

Competency and Practical Skills

This practical is expected to develop the following skills for the industry identified competency 'Maintain microcontroller based systems':

- Select appropriate type of 7-segment display for given application.
- Interface 7-segment display to the microcontroller in static or multiplexed mode.
- Display numbers and characters on 7 segment display.

Relevant Course Outcome(s)

- Interface the memory and I/O devices to 8051 microcontroller.

Practical Outcome

- Interface 7-segment display to display the decimal number from 0 to 9.

Relevant Affective domain related Outcome(s)

- Follow safe practices
- Maintain tools and equipment.
- Follow ethical practices

Minimum Theoretical Background

7-segment display may use a light-emitting diode (LED) for each segment, or other light-generating or controlling techniques such as cold cathode gas discharge, vacuum fluorescent, incandescent filaments, and other. Types of LED based 7-segment display are i) Common Cathode Display ii) Common Anode. Display

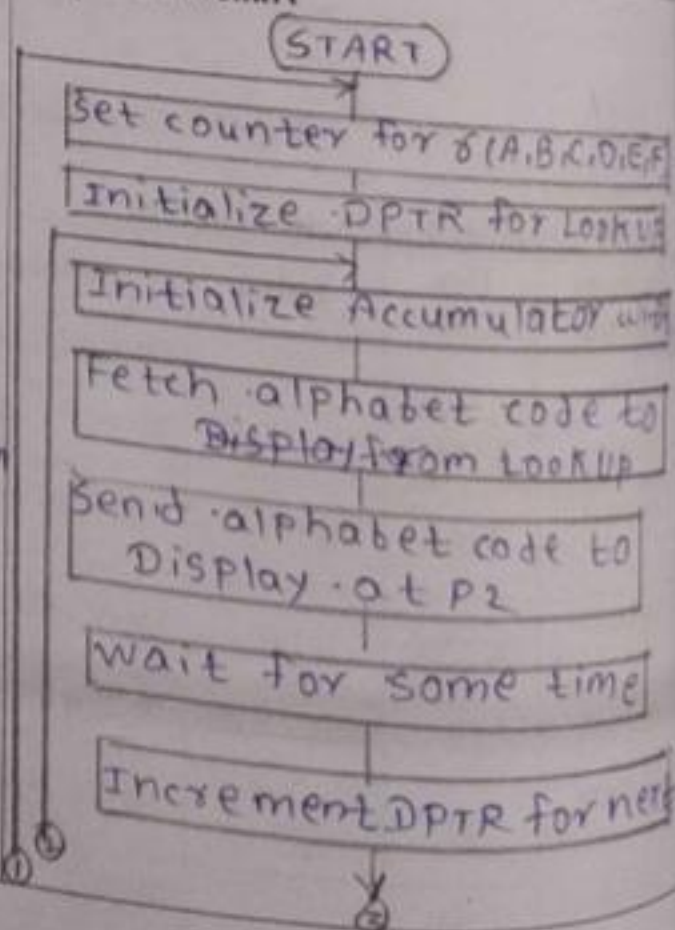
0x000C	1111		ALL DELAY	
0x000E	A3		INC DPTR	:Increment memory pointer to read next digit hex code
C:0x000F	DAF7		DJNZ R2,UP	:Decrement counter & jump if not equal to zero to label UP
C:0x0011	80F0		SJMP REPEAT	:Repeat loop
C:0x0013	7B19	DELAY	MOV R3,#25	:Delay Subroutine
C:0x0015	7C64	L3:	MOV R4,#100	
C:0x0017	7D64	L2:	MOV R5,#100	
C:0x0019	DDEF	L1:	DJNZ R5,L1	
C:0x001B	DCFA		DJNZ R4,L2	
C:0x001D	DBF6		DJNZ R3,L3	
C:0x001F	22		RET	
			ORG 0200H	
		TABLE:	DB 0C0H, 0F9H, 0A4H, 0B0H, 99H, 92H, 82H, 0F8H, 80H, 90H	:Decimal 0 to 9 hex code stored in code memory starting at location 0200H onward
			END	

Problem statement for student: Develop assembly program to display Alphabets A, B, C, D, E and F on Common Anode 7-segment display

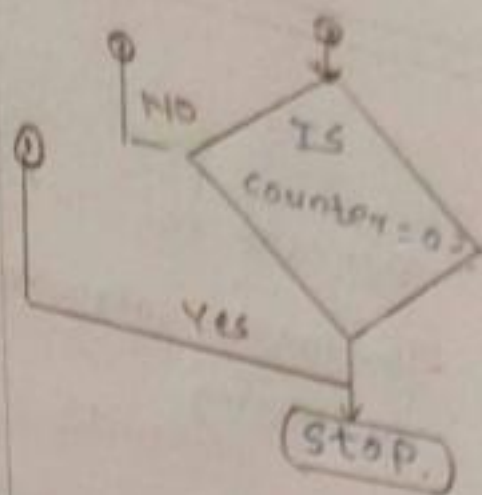
Step 1: Algorithm

- 1) start
- 2) set counter for 6 (A,B,C,D,E,F) Alphabets.
- 3) Initialize DPTR for Look up Table
- 4) Initialize Accumulator with 00H.
- 5) Fetch Alphabet code from Look up Table
- 6) send alphabet code to Display at P2.
- 7) wait for some time.
- 8) Increment DPTR for next Alphabet.

Step 2: Flowchart



3) check counter; if it is not zero, then go to step 4.
 4) Repeat Forever.
 5) stop.



Step 3- Assembly Language Program

Memory Address	Hex Code	Label	Mnemonics	Comments
			ORG 0000H	Initialize
0000H	7806	START:	MOV R0, #6	Set counter for 6 Alphabets
0002H	900100		MOV DPTR, #TABLE	Initialize DPTR for loop up table
0005H	E4	BACK:	CLR A	Initialize Accumulator with 00H
0006H	93		MOVC A, @A+DPTR	Fetch alphabet code from lookup table
0007H	F5A0		MOV P2, A	send alphabet code to display at P2
0009H	1110		ACALL DELAY	wait for some time
000BH	A3		INC DPTR	Increment DPTR for next Alphabet
000CH	D8F7		DJNZ R0, BACK	check counter is zero or not
000EH	80F0		SJMP START	Repeat forever
0010H	7905	DELAY:	MOV R1, #05H	Delay program using registers
0012H	7AFF	UP2:	MOV R2, #0FFH	
0014H	7BFF	UP1:	MOV R3, #0FFH	
0016H	DBFE	UP:	DJNZ R3, UP	
0018H	DAFA		DJNZ R2, UP1	
001AH	D9F6		DJNZ R1, UP2	
001CH	22		RET	Return to main program
			END	

XII Resources Used		Specification	Quantity
S. No.	Instrument / Components		
1.	Desktop / PC	latest processor	1
2.	Simulation software	Keil	1
3.	Interfacing circuits	microcontroller kit, seven segment display.	1

XIII Actual Procedure Followed (use blank sheet provided if space not sufficient)

- 1) write algorithm for given problem.
- 2) Draw flowchart.
- 3) Developed assembly program using Keil IDE.
- 4) Debug program on IDE.
- 5) Execute program on IDE.
- 6) create hex file.
- 7) Download hex code in EPROM/Flash memory.
- 8) Interface CA type 7 segment display to microcontroller as per ckt diagram.
- 9) observe & draw the display of numbers on 7-segment display.
- 10) Record the hex value in observation table.

XIV Precautions Followed (use blank sheet provided if space not sufficient)

- 1) use always current limiting resistor before interfacing 7 segment display to microcontroller.
- 2) For safe operation use seven segment display at 25°C temperature.

XV Observations for sample program (use blank sheet provided if space not sufficient)

perform assembly language program display
alphabet A, B, C, D, E & F on common anode
7 segment display.

Interpretation of Results (Give meaning of the above obtained results)

A seven segment display use a light emitting diode for each segment or other, light generating vacuum & other.

Conclusions and Recommendation

By performing this practical we conclude that develop skill to interface 7 segment display to controller & display number from a tag.

Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design more such questions so as to ensure the achievement of identified CO

1. List the numbers for CC type display in which segment 'a' is activated.
2. Write Steps for testing seven segment displays.
3. Write and execute a program to display continuously '0 to 9' if switch is pressed else '9 to 0' on 7 segment display

[Space for Answers]

→ check all segment of both common cathode & anode to ensure the display it's working properly.

→ Testing seven segment display
i) Hold the display in your hand & Identify pin 1
ii) Now take the multimeter. (Assuming red lead for positive & a black lead for negative) & set it to the proper continuing Range.

iii) check your meter with a sound test (touch the lead together) & a sound will be produce.

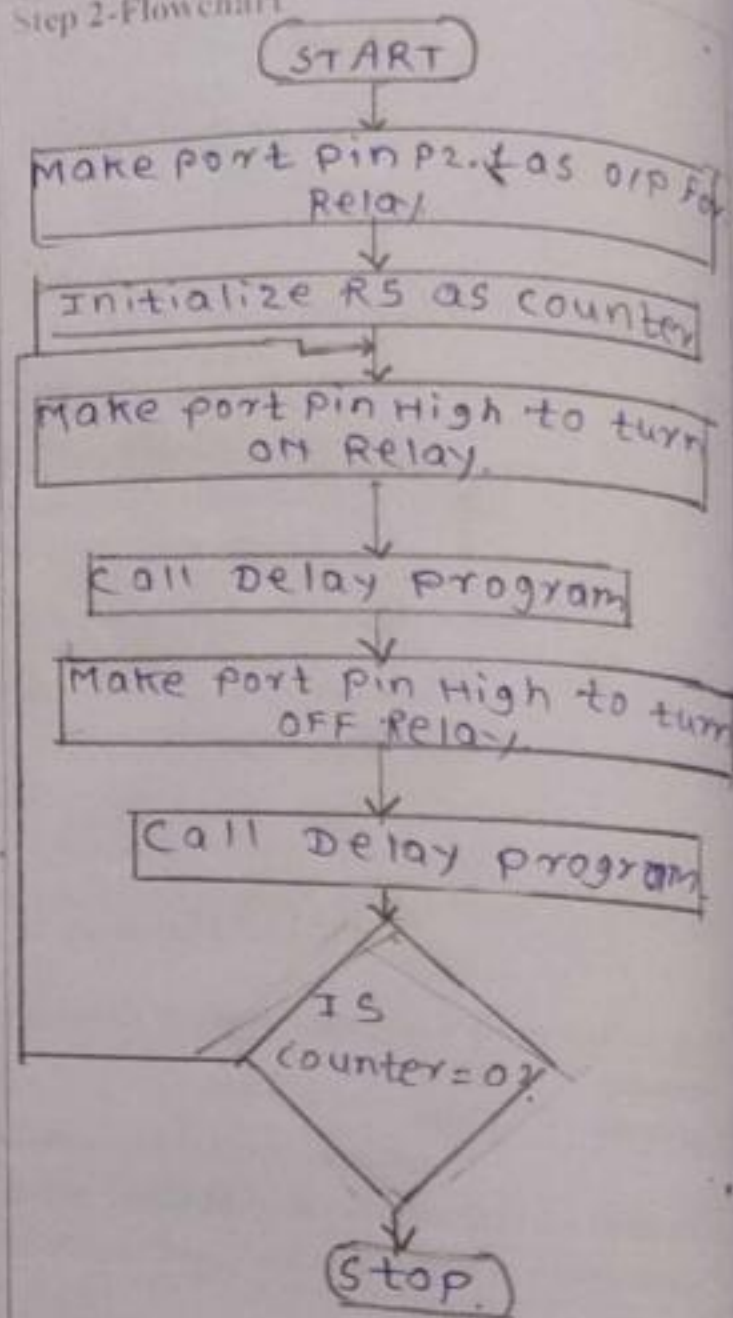
Practical No. 13: Interface relay with microcontroller and turn it ON and OFF

- I **Practical Significance**
In Industrial applications low power devices microcontrollers drive relays are used to control electrical loads beyond their direct drive capability. Relays are used wherever it is necessary to control a high power or high voltage circuit with a low power circuit especially when galvanic isolation is desirable. Electromechanical protective relays are used to detect overload and other faults on electrical lines by opening and closing circuit breakers. This practical will help the students to develop skills to interface relay to microcontroller and turn it ON and OFF.
- II **Relevant Program Outcomes (POs)**
 - **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunications engineering related problems.
 - **Experiments and practice:** Plan to perform experiments and practices to use the results to solve broad-based Electronics and Telecommunication engineering problems.
 - **Engineering tools:** Apply relevant Electronics and Telecommunications technologies and tools with an understanding of the limitations
- III **Competency and Practical Skills**
This practical is expected to develop the following skills for the industry identified competency 'Maintain microcontroller based systems':
 - Select appropriate type of relay for given application.
 - Interface relay to the microcontroller.
 - Use relay to switch on and off electrical loads.
- IV **Relevant Course Outcome(s)**
 - Interface the memory and I/O devices to 8051 microcontroller.
- V **Practical Outcome**
 - Interface relay with microcontroller and turn it ON and OFF.
- VI **Relevant Affective domain related Outcome(s)**
 - Follow safe practices.
 - Demonstrate working as a leader/a team member
 - Maintain tools and equipment..
- VII **Minimum Theoretical Background**
When an electric current is passed through the coil it generates a magnetic field that activates the armature and the consequent movement of the movable contact either makes or breaks (depending upon construction) a connection with a fixed contact.

Step 1-Algorithm

- 1) start.
- 2) Make port pin as output for Relay.
- 3) set counter for 10 times.
- 4) Make port pin High to turn on Relay.
- 5) call delay program.
- 6) Make port pin Low to turn OFF Relay.
- 7) call delay program.
- 8) check counter, if it is zero then goto step 4.
- 9) stop.

Step 2-Flowchart



Memory Address

Maharashtra State Board of Technical Education

XII Resources Used

S. No.	Instrument / Components	Specification	Quantity
1	computer / laptop	latest processor	1
2	simulation software	keil	1
3	interfacing circuits	Relay trainer board.	1

XIII Actual Procedure Followed (use blank sheet provided if space not sufficient)

- 1) Write algorithm for given problem.
- 2) Draw flowchart for the same.
- 3) Develop assembly program on keil IDE.
- 4) Debug program on IDE.
- 5) Execute program on IDE.
- 6) create hex file for above program.
- 7) Download hex code in EPROM / Flash memory.
- 8) Interface relay to the microcontroller as per ckt diagram.
- 9) observe the relay operation by connecting electrical load & note in observation.

XIV Precautions Followed (use blank sheet provided if space not sufficient)

- 1) Use always driver circuit before interfacing relay to the microcontroller.
- 2) Use fly back diode to avoid voltage spikes.

XV Observations for sample program (use blank sheet provided if space not sufficient)

Sr. NO.	Step	Port Pin Status	Logic 1 (+5V) / Logic 0 (0V)	Relay Status ON/OFF
1	Step 1	P2.0		
2	Step 2	P2.0	+5V	ON
			0V	OFF

XVI Results (Output of the Program)

For logic (+ve) then relay status is ON
for logic 0 then relay status is OFF

XVII Interpretation of Results (Give meaning of the above obtained results)
when an electrical circuit is passed through a coil it integrates a magnetic field that activates the consequent movement of the movable contact either makes with a fixed contact.

XVIII Conclusions and Recommendation

By performing this practical we conclude that we develop skills to interfacing relay with microcontroller & turn it off, on.

XIX Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design more such questions so as to ensure the achievement of identified CO

1. Explain what will happen if driver circuit is not connected between microcontrollers and relay.
2. Draw the Interfacing diagram of microcontroller with relay using ULN 2003A IC.
3. Write Steps for testing a relay.

[Space for Answers]

D → If driver circuit is not connected between
i) Relay is an electromagnetic switch which is used to switch high voltage/current using low power circuit relay isolates low power circuit energizing a coil wound on a soft iron core for detailed working of Relay should not be directly connected to a microcontroller it needs a driving circuit.

3 → Steps for testing Relay.

- i) Keep the multimeter in continuity check mode.
- ii) Check for continuity between N/C contact & N/O the pole.
- iii) Check for ^{is} continuity between N/O contact & the pole.

- iv) The Relay will engage with clicking sound.
- v) Now check the continuity between N/O contact.
- vi) Also check for discontinuity betⁿ N/C contact & pole.
- vii) And a final test the measure resistance of the relay coil using a multimeter & check whether it is matching to the voltage printed by manufacturer.

Practical No. 1a: Interface LCD with 8051 microcontroller to display the characters and decimal numbers

I **Practical Significance**
LCDs are used in a wide range of applications including Computers, televisions, instrument panels, calculator's displays. They are common in consumer devices and have replaced cathode ray tube (CRT) displays in most applications. LCDs are interfaced with microcontrollers to display the given integer and character.

II **Relevant Program Outcomes (POs)**

- **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunications engineering related problems.
- **Experiments and practice:** Plan to perform experiments and practices to use the results to solve broad-based Electronics and Telecommunication engineering problems.
- **Lifelong learning:** Engage in independent and life-long learning activities in the context of technological changes also in the Electronics and Telecommunication engineering and allied industry.

III **Competency and Practical Skills**
This practical is expected to develop the following skills for the industry identified competency 'Maintain microcontroller based systems':

- Select appropriate type of LCD for given application.
- Interface LCD display to microcontroller in 4 bit and 8 bit mode.
- Display given integer and character on LCD.

IV **Relevant Course Outcome(s)**

- Interface the memory and I/O devices to 8051 microcontroller.

V **Practical Outcome**

- Interface LCD with 8051 microcontroller to display the characters and decimal numbers.

VI **Relevant Affective domain related Outcome(s)**

- Follow good housekeeping.
- Maintain tools and equipment.
- Follow ethical practices.

VII **Minimum Theoretical Background**
A liquid-crystal display (LCD) is a flat panel display that uses the light modulating properties of liquid crystals. Liquid crystals do not emit light directly. An LCD is of two types depending upon how they are made with either a passive matrix or an active matrix display grid.

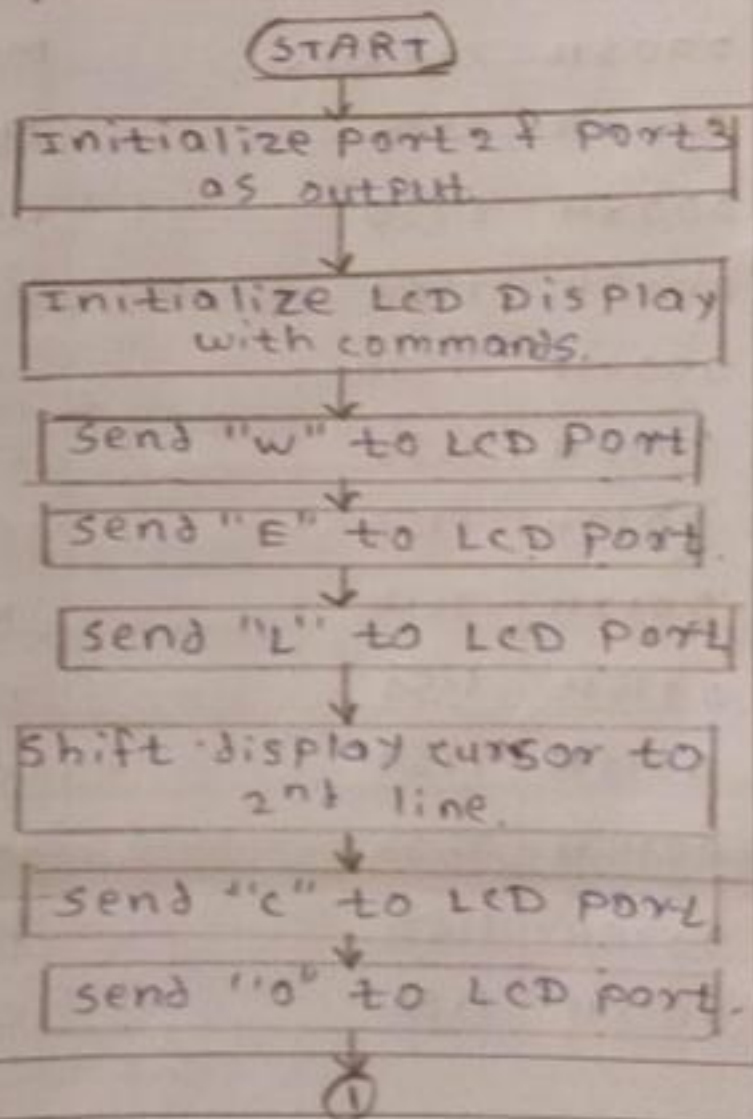
1144		ACALL DELAY	
22		RET	Wait for sometime
1590	LCDDATA:	MOV PLA	Send data to led
02 B3		SETB RS	Select data register
C2 B4		CLR RW	Select write operation
D2 B5		SETB EN	
C2 B5		CLR EN	Latch data to led
1144		ACALL DELAY	Wait for sometime
22		RET	
7B32	DELAY:	MOV R3,#50	Delay subroutine
7CFF	LOOP2:	MOV R4,#255	
DCFE	LOOP1:	DJNZ R4,LOOP1	
DBFA		DJNZ R3,LOOP2	
22		RET	
		ORG 0050H	
	MSG:	DB "AT89C51"	Define data byte to code memory
		END	

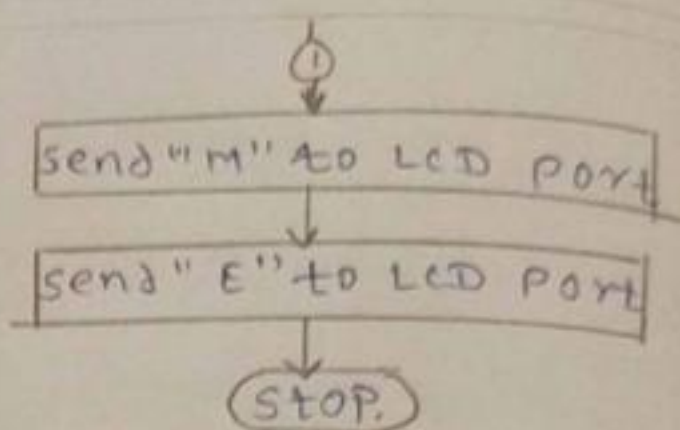
Problem statement for student: Develop assembly program to display data "Wel" on 1st line and "Come" on 2nd line of LCD

Step 1-Algorithm

- 1) Start.
- 2) Initialize port 2 & port 3 as output.
- 3) Initialize LCD Display with commands.
- 4) SEND "W" to LCD port.
- 5) SEND "E" to LCD port.
- 6) Send "L" to LCD port.
- 7) shift display cursor to 2nd line.
- 8) send "c" to LCD port.
- 9) send "o" to LCD port.
- 10) send "m" to LCD port.
- 11) send "E" to LCD port.
- 12) Stop.

Step 2-Flowchart





Step 3- Assembly Language Program

Memory Address	Hex Code	Label	Mnemonics	Comments
			ORG 0000H	Initialize
0000H	75B000		MOV P3, #00H	Declare P2 & P3 as O/P.
0003H	75A000		MOV P2, #00H	
0006H	7438		MOV A, #38H	display 2 line 5x7 matrix 8 bit
0008H	1150		CALL LCD CMD	
000AH	116A		CALL DELAY	
000CH	740E		MOV A, #0EH	cursor on Display on
000EH	1150		CALL LCD CMD	
0010H	116A		CALL DELAY	
0012H	7414		MOV A, #14H	Right Shift cursor position
0014H	1150		CALL LCD CMD	
0016H	116A		CALL DELAY	
0018H	7457		MOV A, # "W"	Send "W" to LCD port
001AH	115D		ACALL LCD DATA	
001CH	116A		CALL DELAY	

001EH	7445		MOV A, # "E"	Send "E" to LCD port.
0020H	115D		ACALL LCD DATA	
0022H	116A		CALL DELAY	
0024H	744C		MOV A, # "L"	Send "L" to LCD port.
0026H	115D		ACALL LCD DATA	
0028H	116A		CALL DELAY	
002AH	74C0		MOV A, # 0C0H	command to start display at 2nd line.
002CH	1150		ACALL LCD CMD	
002EH	116A		CALL DELAY	
0030H	7414		MOV A, # 14H	shift cursor position right.
0032H	1150		ACALL LCD CMD	
0034H	116A		CALL DELAY	
0036H	7443		MOV A, # "c"	Send "c" to LCD port.
0038H	115D		ACALL LCD DATA	
003AH	116A		CALL DELAY	
003CH	744F		MOV A, # "0"	Send "0" to LCD port.
003EH	115D		ACALL LCD DATA	
0040H	116A		CALL DELAY	
0042H	744D		MOV A, # "M"	Send "M" to LCD port.
0044H	115D		ACALL LCD DATA	
0046H	116A		CALL DELAY	
0048H	7445		MOV A, # "E"	Send "E" to LCD port.
004AH	115D		ACALL LCD DATA	
004CH	116A		CALL DELAY	
004EH	80FE		SJMP \$	Stay Here.
0050H	F5A0	LCD CMD:	MOV P2, A	program to send command to LCD.
0052H	C2B3		CLR P3.3	
0054H	C2B4		CLR P3.4	
0056H	D2B5		SETB P3.5	
0058H	116A		CALL DELAY	
005AH	C2B5		CLR P3.5	
005CH	22		RET	

Memory Address	Hex code	LABEL	Mnemonics	Comments
005DH	F5A0	LCD DATA:	MOV P2, A	program to send data to lcd
005FH	D2B3		SETB P3.3	
0061H	C2B4		CLR P3.4	
0063H	D2B5		SETB P3.5	
0065H	116A		CALL DELAY	
0067H	C2B5		CLR P3.5	
0069H	22		RET	
006AH	7BFF	DELAY:	MOV R3, #0FFH	Delay program
006CH	7CFF	UP1:	MOV R4, #0FFH	
006EH	DCFE	UP:	DJNZ R4, UP	
0070H	DBFA		DJNZ R3, UP1	
0072H	DBF		RET	
			END	

XII Resources Used

S. No.	Instrument / Components	Specification	Quantity
1.	Desktop PC	Latest processor	1
2.	Simulation software	Keil	1
3.	Interfacing circuit		1.

XIII Actual Procedure Followed (use blank sheet provided if space not sufficient)

- 1) Write algorithm for given program.
- 2) Draw flowchart for the same.
- 3) Develop assembly program using Keil IDE.
- 4) Debug program on IDE.
- 5) Execute program on IDE.
- 6) Create hex file for the above program.
- 7) Interface LCD display to microcontroller.
- 8) Download hex code in EPROM/Flash memory.
- 9) Observe output on LCD display & note it in observation table.

XIV Precautions Followed (use blank sheet provided if space not sufficient)

- 1) LCD display made up of glass avoid applying strong pressure on to the surface of display area.
- 2) Ensure proper connection then only give electric supply to circuit.

XV Observations for sample program (use blank sheet provided if space not sufficient)

Sr. No.	LCD Memory Location	Observed Alphabet and decimal number
1	84H	
2	85H	M
3	86H	S
4	87H	B
5	88H	T
6		E
7		
8		
9		
10		

Results (Output of the Program)

The code is displayed on LCD display on ~~code~~ ^{LCD} is 'MSB7E'. The code

Interpretation of Results (Give meaning of the above obtained results)

The LCD is off two modes with either a passive matrix or a active matrix display grid.

Conclusions and Recommendation

LCD are interfaced with micro controller to display the given integer.

Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design more such questions so as to ensure the achievement of identified CO

1. Write down cost of LCD module for at least 3 different Manufacturer with specifications.
2. State possible reasons of 16x2 LCD display not working
3. Write assembly program for checking busy flag in LCD display.
4. Write appropriate command to shift entire data to right.

[Space for Answers]

- 2) → i) Display showing black boxes
ii) Display showing wrong character
iii) Display showing black lines

4) → select Full length or line that want shift to move the text to the right
press tab to move the text to right
right click edit > shift register

Practical No. 12: Interface DAC with 8051 microcontroller and observe following waveforms: square wave, triangular wave, sawtooth wave

I Practical Significance

The digital to analog converter (DAC) is a device which used to convert digital signals into analog signals. This practical will help the students to develop skills to interface DAC with 8051 and generate different analog waveforms.

II Relevant Program Outcomes (POs)

- **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunication engineering related problems.
- **Experiments and practice:** Plan to perform experiments and practices to use the results to solve broad-based Electronics and Telecommunication engineering problems.
- **Engineering tools:** Apply relevant Electronics and Telecommunication technologies and tools with an understanding of the limitations.
- **Lifelong learning:** Engage in independent and life-long learning activities in the context of technological changes also in the Electronics and Telecommunication engineering and allied industry.

III Competency and Practical Skills

This practical is expected to develop the following skills for the industry identified competency 'Maintain microcontroller based systems':

- Select appropriate type DAC for given application
- Interface DAC with microcontroller.
- Generate different analog signals.

IV Relevant Course Outcome(s)

- Maintain microcontroller used in different application.

V Practical Outcome

- Interface DAC with 8051 microcontroller and observe following waveforms: square wave, triangular wave, sawtooth wave.

VI Relevant Affective domain related Outcome(s)

- Follow safe practices.
- Practice good housekeeping.
- Practice energy conservation
- Maintain tools and equipment

VII Minimum Theoretical Background

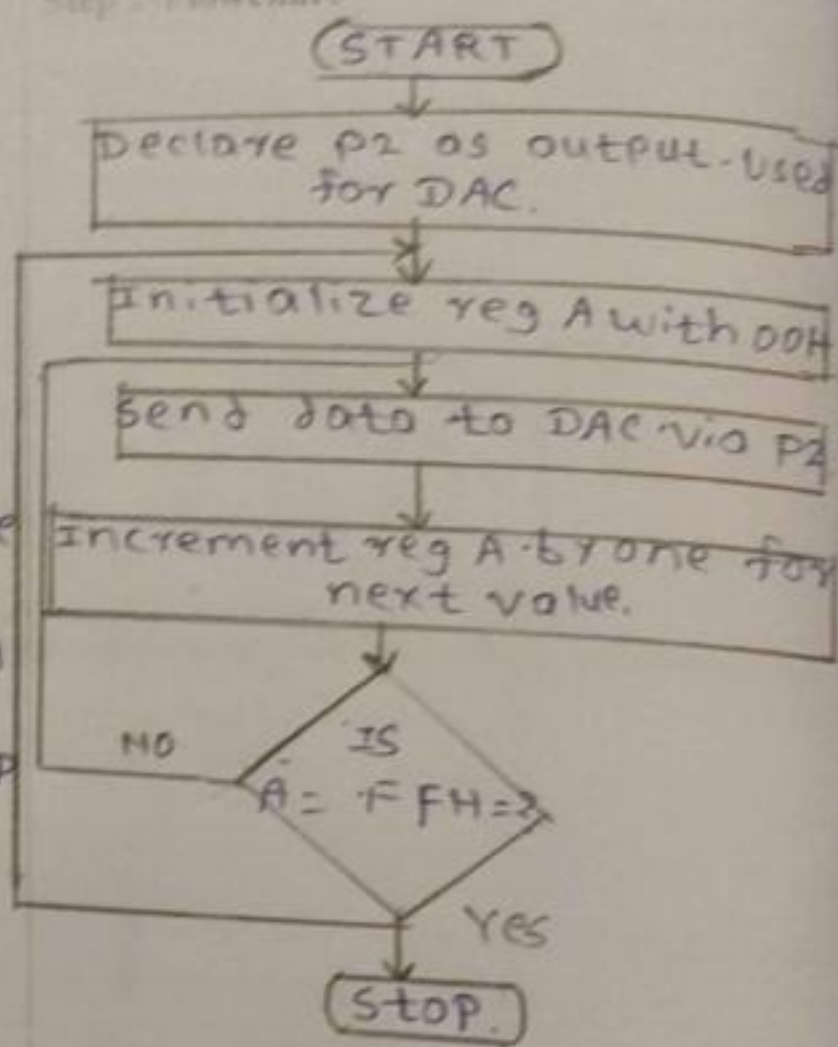
In DAC the number of data bit input decides the resolution since the number of analog output levels is equal to 2^n , where n is the number of data bit inputs. An 8 input DAC provides 256 discrete voltage (or current). The most commonly used, 8 bit, R/2R method followed DAC is DAC 0808

Problem statement for an 8085 microprocessor program to generate square wave using DAC 0804

Step 1-Algorithm

- 1) start.
- 2) Declare p2 as output used for DAC.
- 3) initialize reg A with 00H.
- 4) send data to DAC via p2.
- 5) increment reg A by one for next value.
- 6) If A not equal to FFH then goto step 4 otherwise goto step 3 to repeat.
- 7) stop.

Step 2-Flowchart



Step 2: Assembly Language Program

Memory Address	Hex Code	Label	Mnemonics	Comments
			ORG 0000H	Initialize
0000H	75A000		MOV P2, #00H	Declare P2 as o/p used for DAC
0003H	7400	START:	MOV A, #00H	Initialize Reg A with 00H
0005H	F5A0	BACK:	MOV P2, A	Send data to DAC via P2.
0007H	04		INC A	Increment Reg A by one - to send next value of P2.
0008H	B5FFFA		CJNE A, 0FFH, BACK	If A not equal to FFH then go back
000BH	80F6		SJMP START	If A equal to FFH then go to START - to repeat
			END	

XII Resources Used

S. No.	Instrument / Components	Specification	Quantity
1.	Dekstop / PC	loaded with simulation software	1
2.		keil	1
3.	software	DAC(0808) trainer	1
4.	Interfacing circuit	Board.	

XIII Actual Procedure Followed (use blank sheet provided if space not sufficient)

- 1) Write algorithm for given problem.
- 2) Draw flowchart for the same.
- 3) Develop assembly program using Keil IDE.
- 4) Debug program on IDE.
- 5) Execute program on IDE.
- 6) Create hex file for the above program.
- 7) Download hex code in EPROM/Flash memory.
- 8) Interface DAC 0808 IC to the microcontroller.
- 9) Observe triangular waveform on CRO.

XIV Precautions Followed (use blank sheet provided if space not sufficient)

- 1) Operate DAC chip as per specifications given in the datasheet otherwise damage may occur to the device.

Step 2: Assembly Language Program

Memory Address	Hex Code	Label	Mnemonics	Comments
			ORG 0000H	Initialize
0000H	75A000		MOV P2, #00H	Declare P2 as O/P used for DAC
0003H	7400	START:	MOV A, #00H	Initialize Reg A with 00H
0005H	F5A0	BACK:	MOV P2, A	Send data to DAC via P2.
0007H	04		INC A	Increment Reg A by one - to send next value of P2.
0008H	B5FFFA		CJNE A, 0FFH, BACK	If A not equal to FFH then go back
000BH	80F6		SJMP START	If A equal to FFH then go to START - to repeat
			END	

XII Resources Used

S. No.	Instrument / Components	Specification	Quantity
1.	Desktop / PC	loaded with simulation software	1
2.		Keil	1
3.	software	DAC (0808) trainer	1
4.	Interfacing circuit	Board	

XIII Actual Procedure Followed (use blank sheet provided if space not sufficient)

- 1) Write algorithm for given problem.
- 2) Draw flowchart for the same.
- 3) Develop assembly program using Keil IDE.
- 4) Debug program on IDE.
- 5) Execute program on IDE.
- 6) Create hex file for the above program.
- 7) Download hex code in EPROM / Flash memory.
- 8) Interface DAC 0808 IC to the microcontroller.
- 9) Observe triangular waveform on CRO.

XIV Precautions Followed (use blank sheet provided if space not sufficient)

- 1) Operate DAC chip as per specifications given in the datasheet otherwise damage may occur to the device.

XXI Results (Output of the Program)

When we apply digital coded signal to DAC converter it converts the digital signal into analog triangular waveform by creating loop in the program.

XXII Interpretation of Results (Give meaning of the above obtained results)

In DAC, no. of data bit input decides the resolution since the no. analog output level is equal to 2^n where n is the no. of data bits input.

XXIII Conclusions and Recommendation

By performing this practical we have learned & observed the triangular wave by using interface DAC with 8051 microcontroller.

XXIV Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design more such questions so as to ensure the achievement of identified CO

1. If $I_{ref} = 2\text{mA}$ and all the inputs to the DAC are high then find maximum current of DAC 0808 IC.
2. To generate a sine wave using DAC 0808 find decimal values representing magnitude of the sine of angles between 0 and 360 degrees.
Refer $V_{out} = 5V + (5 \times \sin\theta)$.
3. If a switch SW is connected to port pin P0.0. Write a program to do the following.
 - a) When $SW=0$ the DAC output gives a triangular waveform.
 - b) When $SW=1$ the DAC output gives a staircase waveform.

[Space for Answers]

2) → To generate a sinewave using DAC0808
Sine wave angle is as

i) 260°

ii) 60°

iii) 90°

iv) 180°

v) 360°

For 60°

$$\begin{aligned}V_{out} &= 5 + [5 \sin 60^\circ] \\&= 5.33 \times 0.560 \\&= 2.38 \text{ d.}\end{aligned}$$

$$\begin{aligned}\text{for } 90^\circ &= 5 + [5 \sin 90^\circ] \\&= 10 \times 25.60 \\&= 250 \text{ d.}\end{aligned}$$

1) → If $I_{ref} = 2 \text{ mA}$

on input to DAC is high the max^m
current of DAC 0808 IC is 1.99 mA

Practical No. 18: Interface stepper motor to microcontroller and rotate in clockwise and anti-clockwise direction at the given angles.

I Practical Significance

Different field applications require precise positioning, repeatability of movement in clockwise and anticlockwise direction with good accuracy. Stepper motors are controlled by microcontrollers in such areas like in computer peripherals, Business machines, process control and for making robots. This practical will help the students to develop skills to interface stepper motor to 8051 and rotate in clockwise and anticlockwise direction.

II Relevant Program Outcomes (POs)

- **Discipline knowledge:** Apply Electronics and Telecommunication engineering knowledge to solve broad-based Electronics and Telecommunications engineering related problems.
- **Experiments and practice:** Plan to perform experiments and practices to use the results to solve broad-based Electronics and Telecommunication engineering problems.
- **Engineering tools:** Apply relevant Electronics and Telecommunications technologies and tools with an understanding of the limitations.

III Competency and Practical Skills

This practical is expected to develop the following skills for the industry identified competency 'Maintain microcontroller based systems':

- Select appropriate type of stepper motor for given application.
- Interface stepper motor with microcontroller.
- Apply full step or half step sequence as per requirement of movement.

IV Relevant Course Outcome(s)

- Maintain microcontroller used in different application.

V Practical Outcome

- Interface stepper motor to microcontroller and rotate in clockwise and anti-clockwise direction at the given angles.

VI Relevant Affective domain related Outcome(s)

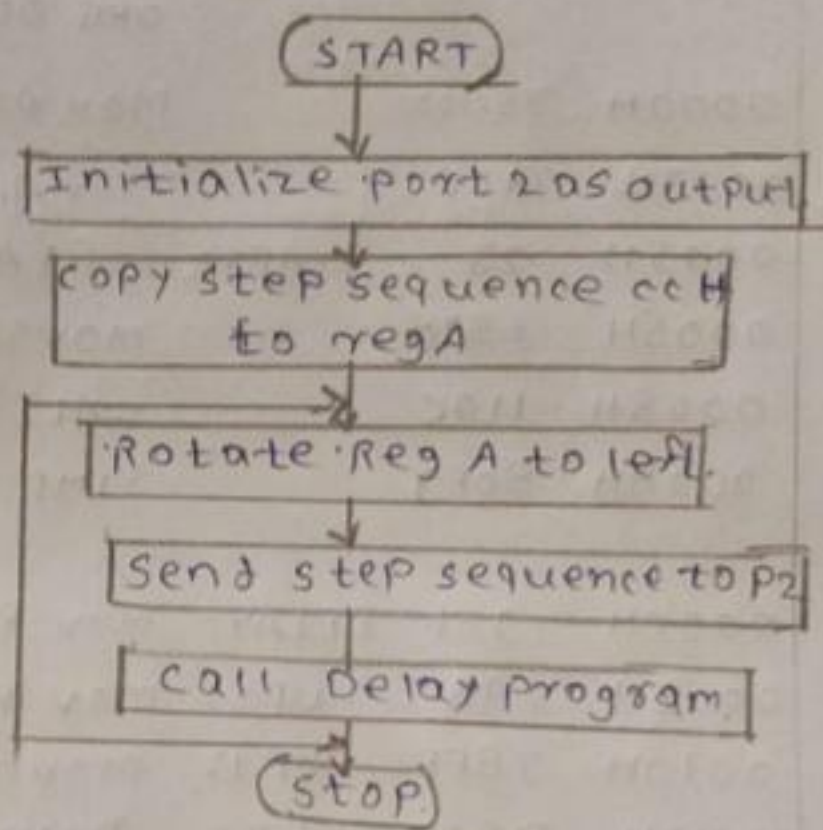
- Follow safe practices.
- Maintain tools and equipment.
- Follow ethical practices.

problem statement for student: Develop assembly program for stepper motor to rotate 90 anticlockwise direction using rotate instruction. Assume step angle 1.8°

Step 1 - Algorithm

- 1) start.
- 2) initialize port 2 as output for stepper motor interfacing.
- 3) copy step sequence cch to reg A.
- 4) Rotate Reg A to left for anticlockwise rotation of stepper motor.
- 5) send step sequence of P₂.
- 6) call delay program.
- 7) Go back to step 4 to Repeat.
- 8) stop.

Step 2 - Flowchart



Step 3: Assembly Language Program

Comments

Memory
AddressHex
Code

Label

Mnemonic

ORG 0000H

0000H	75A500		MOV P1, #00H	Delay P1.2 as input
0003H	74CC		MOV A, #00CH	Make P1.2 High
0005H	23	START:	RLA	call Delay of 10ms
0006H	F5A0		MOV P1, A	Make P1.2 Low
0008H	110C		CALL DELAY	call Delay of 10ms
000AH	80F3		SJMP START	Repeat Forever
000CH	7901	DELAY:	MOV R1, #01H	Delay Program
000EH	7AFF	UP1:	MOV R2, #0FFH	
0010H	78FF	UP1:	MOV R3, #0FFH	
0012H	DBFE	UP:	DJNZ R3, UP1	
0014H	DAFA		DJNZ R2, UP1	
0016H	D9F6		DJNZ R1, UP2	
0018H	22		RET	Return to main program
			END	

No.	Instrument Components	Specification	Quantity
1	computer system	Loaded with IDE keil.	1
2	Microcontroller kit	single board system	1
3	Stepper motor		
4	Trainer.	1.8° step angle, 50/100 RPM	1

XIII Actual Procedure Followed (use blank sheet provided if space not sufficient)

- 1) Write algorithm for given program.
- 2) Draw flowchart for the same.
- 3) Develop assembly program using Keil IDE.
- 4) Debug program on IDE.
- 5) Execute program on IDE.
- 6) Create hex file for above program.
- 7) Download hex code in EPROM/Flash memory.
- 8) Interface stepper motor to microcontroller.
- 9) Observe rotation of stepper motor & record observation.

XIV Precautions Followed (use blank sheet provided if space not sufficient)

1. Use always driver circuit while interfacing stepper motor to microcontroller.

XV Observations for sample program (use blank sheet provided if space not sufficient)

Step Code	Port Pin	Status - logic 1 (+5V) / logic 0 (0V)
9H	P2.4 - P2.0	
CH	P2.4 - P2.0	
6H	P2.4 - P2.0	
3H	P2.4 - P2.0	

XVI Results (Output of the Program)

XVII Interpretation of Results (Give meaning of the)

By initializing port 2 as output, copy sequence reg A as per step sequence rotated the motor clockwise or anticlockwise.

XVIII Conclusions and Recommendation

By performing this practical we conclude to develop assembly program for stepper motor to rotate 30° in anticlockwise direction using rotate instruction.

XIX Practical Related Questions

Note: Below given are few sample questions for reference. Teacher must design more such questions so as to ensure the achievement of identified CO

1. Write specifications of 4 different stepper motors available in the market.
2. Calculate no of steps to move stepper motor 64 degree in clockwise direction if step angle is 2 degree, steps per revolution 180 & no of rotor teeth 45.
3. If a motor takes 90 steps to make one complete revolution, what is the step angle of this motor?

[Space for Answers]

- 1) $\rightarrow$ Name: Intelligent Nema 34 stepper motor
Voltage: 24-48 Vdc
Model no: - 86A8
Torque: 12.7 mm
Rated current: 4.9 A
- 2) Name: Tin can stepper motor
Weight: 300 gm
Bipolar operation: 17.1 Ncm
Electrical enclosure: 407p
Steps per revolution: 48
- 3) Name: Nema 23 closed loop stepper motor
No. of phase: 2
Voltage: $< 100V$
Power source: DC
Model number: 17HS4401

$$\begin{aligned}
 3) \rightarrow 90 \text{ steps} &= 360 \\
 1 \text{ step} &= \frac{360}{90} \\
 &= 4
 \end{aligned}$$

XX References / Suggestions for further reading

1. https://en.wikipedia.org/wiki/Stepper_motor.
2. <https://www.jameco.com/Jameco/Products/ProdDS/155433.pdf>
3. The 8051 Microcontroller and Embedded system Using Assembly and C- Muhammad Ali Mazidi, Janice Gillispie Mazidi, Rolin D. Mckinlay- Pearson /Prentice Hall, . 2nd edition, Delhi, 2008, ISBN 978-8177589030
4. <https://www.youtube.com/watch?v=TWMai3oimM>